Prediction maps as a gateway to better understand and test scientific theories

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Abstract

Scientific theories aim to explain and predict phenomena, yet assessing how experimental evidence impacts a theory remains a significant challenge. This is particularly evident in the psychological and cognitive sciences, where a lack of formal models and precise quantitative predictions often obscures the "centrality" of a given hypothesis for the theory that generates it. Because auxiliary hypotheses are often omitted or vaguely related to core ideas of a theory, researchers can easily safeguard core theoretical tenets against falsified predictions, rendering theory testing largely ineffective. To address this problem, we introduce prediction maps: a novel methodological approach that represents scientific theories as networks of beliefs structured in a core-periphery way. They visualize theoretical claims and empirical predictions as nodes, with inferential relationships as edges. By applying graph-theory and network analysis metrics, we can quantify the centrality of specific predictions. This allows for a systematic evaluation of "evidential weight," identifying which experimental results would be the most meaningful tests for a theory. We demonstrate the utility of this approach within the science of consciousness – a field characterized by high theoretical fragmentation and underspecified models. We show how prediction maps can be used to design more informative experiments, compare competing frameworks, and assess empirical success. Finally, we present an open-access online platform designed to help researchers across disciplines structure their theories. Prediction maps offer a transparent, structural analysis of the link between theory and evidence, providing a rigorous tool for navigating the complex landscape of theory testing.

Keywords: Theory testing; prediction maps; centrality; scientific theories; consciousness

Introduction

One of the central aims of scientific theories is to explain and predict true states of affairs (Douglas, 2009; Hofstadter, 1951; Salmon, 1978), with predictive accuracy often considered the hallmark of good scientific theories (Kuhn, 1981; McMullin, 1982). Consequently, theory testing is one of the central aspects of scientific practice. In the most common method for theory testing, termed "the hypothetico-deductive method" (Sprenger, 2011), predictions are extracted from theories, and tested experimentally in order to assess their strength. Yet, determining whether experimental evidence accords with a theory's predictions, and to what extent, is not always straightforward (Crupi, 2021; Howson & Urbach, 1989; Sprenger, 2011). Indeed, although statistical methods allow testing whether the experimental data fit with a specific hypothesis, it is often unclear how the results reflect on the theory from which the hypothesis was derived. That is, refuting hypotheses that are central to a given theory should challenge the theory more than refuting less central predictions (Grim et al., 2022). The key problem is that it is not always clear which prediction is central and which is not, providing theory leaders substantial leeway in safeguarding their theory in the case of falsified hypotheses (Lakatos, 1976).

This is particularly true in the psychological and cognitive sciences, where operationalization of theoretical constructs often requires many assumptions, and commonly does not involve formal models that enable precise quantitative predictions (Eronen & Bringmann, 2021; Guest & Martin, 2021; Meehl, 1967, 1978; Oberauer & Lewandowsky, 2019; Van Rooij & Blokpoel, 2020). While progress has been made in improving methods for hypothesis testing (i.e., quantifying the extent to which experimental data fit a hypothesis, see Cumming, 2014; Morey et al., 2014; Yarkoni & Westfall, 2017), there is often insufficient clarity about the centrality of the hypothesis to the theory generating it. The lack of formal models to mediate theory and evidence makes it particularly difficult to assess the meaningfulness of a test for a given theory, contributing to the "theory crisis" in psychology (Oberauer & Lewandowsky, 2019; see also Eronen & Bringmann, 2021). To allow for theory assessment, empirical testing must be paired with an analysis that clarifies the impact of any piece of evidence on a theory. This requires elucidating the logical links between theoretical constructs and testable predictions.

Here, we adopt a novel approach for doing so, thereby advancing theory testing: "prediction maps", which are representations of scientific theories as networks of beliefs (Grim et al., 2022;

Henderson et al., 2010; Quine, 1951). The different claims and empirically testable predictions made by the theories are represented as nodes of the network, while the edges between the nodes represent the logical relation between the theory's claims.

This approach follows a Lakatosian framework (Lakatos, 1976), which considers scientific theories as networks of beliefs centered around core principles and various levels of peripheral auxiliary hypotheses. These auxiliary hypotheses render the core ideas empirically testable, while also shielding them from direct falsification: for every failed prediction, the theorist can reject the auxiliary hypothesis that made the empirical prediction possible, while holding on to the core idea on which the prediction was based. Prediction maps articulate these hypotheses, as well as their relation to the core. In this way, these maps can be used to navigate the complex relationship between core principles and observable consequences of a theory, offering a novel methodology for systematically assessing the relevance of experimental data for a general theory or framework.

Our approach assumes that (dis)confirmatory relationships between theoretical constructs and empirical evidence can be represented as logical relationships between statements (for criticism, see Bogen & Woodward, 2005; for a reply, see Votsis, 2011); though we do not require that these relationships be deductive (our framework is thus compatible both with the standard hypothetico-deductive method and the inductive methodology of Bayesian confirmation theory). In virtue of these logical connections, falsifying an empirical statement jeopardizes the integrity of the network (Feigl, 1970; Grim et al., 2022; Henderson et al., 2010; for general discussion on the distinction between theoretical and observation statements, see Boyd & Bogen, 2026). This implies that, depending on the structure of the theory and the interconnectivity among its claims, refuting specific hypotheses amounts to damaging certain "areas" of the theory more than others: the more central and interconnected a prediction is, the more difficult it should be to amend the overall framework in light of disconfirmatory evidence.

Visualizing scientific theories as prediction maps centered around some core principles allows one to identify more central claims and predictions that are particularly influential to the theory. The centrality, or importance, of each claim can be calculated using graph-theory and network analysis metrics (Borgatti & Everett, 2006) and used as a proxy for evaluating the evidential weight that a given piece of evidence has over a theory. This, in turn, provides a more informed

framework for assessing the meaning of existing data, as well as novel experimental designs, for evaluating the theory. For example, researchers can use prediction maps to decide which claim is better to target when designing an experiment (e.g., by determining which prediction would be more "disruptive" to the theory), to compare similarities and dissimilarities between theories, and to assess their empirical success.

In this paper, we demonstrate how to construct prediction maps for scientific theories, using the science of consciousness as a case in point. We further provide an online platform (accessible here: **https://xk29y8.csb.app/**) that allows researchers of all disciplines to create new maps for theories in their field, and show – using this case study – how this could be a major tool for advancing theory testing and understanding. The study of consciousness is especially suited for this purpose, as it is populated by many different theories, all trying to account for how the brain relates to conscious experience (Seth & Bayne, 2022). These theories are quite heterogeneous, indicating that consciousness science has not yet found a dominant paradigm (Kuhn, 1970). This theoretical fragmentation is partly due to the fact that most theories are still underspecified frameworks, rather than rigorous and formalized models (Mudrik et al., 2025; Signorelli et al., 2021). Moreover, different theories often make incompatible assumptions about the nature of consciousness (Mudrik et al., 2025; Seth & Bayne, 2022). As a result, they develop by following different research questions, with some methods being more prone to confirm specific hypotheses (Yaron et al., 2022). When the relationship between theory and evidence is not constrained by strict logical relations and quantitative predictions, theory evaluation is particularly challenging (Chis-Ciure et al., 2024; Fazekas et al., 2024; Kirkeby-Hinrup, 2024; Negro, 2024). This renders prediction maps especially useful, as they pair empirical evidence with a conceptual and structural analysis of the theory and the informative value of its predictions.

Notably, prediction maps are, in essence, *models of models:* they are idealized, abstracted, and simplified representations of scientific theories, much like scientific models are idealized representations of a target phenomenon (Bokulich, 2011; Chirimuuta, 2024). This implies that the particular way a theory is represented as a prediction map will partially depend on the choices and epistemic aims of the modeler (Giere, 2004; Massimi, 2018). Building the maps involves some interpretation, which inevitably entails subjective judgement. Here, we tried to

minimize such judgement by aiming for inter-modeler agreement and by conferring with theory proponents to make sure the maps accurately represent their theories.

Methods

Selection of theories

As a starting point and a proof of concept for this approach, we selected four prominent neuroscientific theories of consciousness: the Global Neuronal Workspace Theory (GNWT), Integrated Information Theory (IIT), Perceptual Reality Monitoring (PRM), and Recurrent Processing Theory (RPT). Though there are many other theories that aim at explaining consciousness, these four are often referred to as highly influential (Cleeremans et al., 2025; Del Pin et al., 2021; Doerig et al., 2021; Lau & Rosenthal, 2011; Mudrik et al., 2025; Northoff & Lamme, 2020; Seth & Bayne, 2022; Storm et al., 2024; Yaron et al., 2022), with relatively extensive literature available on each. An exception is PRM, which is a fairly young theory that has been systematically formulated only in recent years (Lau, 2022; Michel, 2023); however, it belongs to a family of theories called "higher-order theories" (HOT; see Brown et al., 2019 for a review) that have emerged as empirical paradigms in the past couple of decades, whose philosophical lineage traces back to Locke and Kant (Carruthers & Gennaro, 2001). Here, we selected PRM as the prime example of HOT because it currently seems to be the most comprehensive *empirical* version of HOT.

Following the creation of these maps, we worked with proponents of additional theories (Brown, 2015; Cleeremans et al., 2020; Fleming, 2020; Pennartz, 2022; Whyte et al., 2026), all tested under adversarial collaboration initiatives (Potgieter, 2024), to create additional maps. As these were not established as part of this work, we include them in the supplementary materials (Supplementary S3), for interested readers.

Selection of papers

To construct a prediction map for each of the four theories, we first read several foundational papers for each theory (for GNWT: Dehaene & Naccache, 2001; Mashour et al., 2020; for IIT: Albantakis et al., 2023; Tononi et al., 2016; for PRM: Lau & Rosenthal, 2011; Brown et al., 2019; for RPT: Lamme, 2006, 2010, 2020). From this literature, we derived further papers about

relevant experimental studies designed to validate specific predictions of the respective theory (see Table 1 for the comprehensive bibliography for each theory).

Theory	*Bibliography*
GNWT	(1. Dehaene & Naccache, 2001; 2. Naccache et al., 2002; 3. Dehaene et al., 2003; 4. Dehaene & Changeux, 2005; 5. Dehaene et al., 2006; 6. Bekinschtein et al., 2009; 7. Sackur & Dehaene, 2009; 8. Dehaene, 2008; 9. Dehaene & Changeux, 2011; 10. Sergent et al., 2013; 11. Salti et al., 2015; 12. Trübutschek et al., 2017; 13. Dehaene et al., 2017; 14. Mashour et al., 2020; 15. Mudrik et al., 2024; 16. Cogitate et al., 2025; 17. Klatzmann et al., 2025).
IIT	(1. Tononi et al., 2016; 2. Tsuchiya et al., 2016; 3. Hoel et al., 2016; 4. Boly et al., 2017; 5. Massimini & Tononi, 2018; 6. Haun & Tononi, 2019; 7. Grasso et al., 2021; 8. Albantakis et al., 2023; 9. Tononi et al., 2024; 10. Olcese et al., 2024; 11. Boly et al., 2024; 12. Mudrik et al., 2024; 13. Cogitate et al., 2025; 14. Haun & Tononi, 2025; 15. Comolatti et al., 2025)
PRM	(1. Lau & Rosenthal, 2011; 2. Odegaard et al., 2017, 3. 2018; 4. Knotts et al., 2019; 5. Brown et al., 2019; 6. Lau, 2022; 7. Butlin et al., 2023; 8. Michel, 2023; 9. Fleming et al., 2025)
RPT	(1. Roelfsema, 2006; 2. Lamme, 2006, 3. 2010; 4. Vandenbroucke et al., 2011; 5. Fahrenfort et al., 2017; 6. Lamme, 2018, 7. 2020; 8. Butlin et al., 2023)

Table 1. *References for each theory*

Extraction of claims and structuring of the maps

We extracted critical theoretical claims and empirical predictions, ensuring that the statements constituting each prediction map adhered as much as possible to the formulation found in the original publications. After the initial extraction, an iterative process of revisions and discussions began, where the claims were placed on the maps, and additional theoretical and experimental papers were added, until a consensus was reached. This process was also aimed at minimizing subjectivity in designing the maps.

We structured each map around a core claim, representing the principal idea of the theory for explaining consciousness. From this core claim stem five different levels of auxiliary hypotheses,

with each level further specifying the information present in the previous level (Henderson et al., 2010). Thus, the first level after the core (L1) unpacks the concepts in the core, the second level (L2) develops the concepts in L1, and so on (Figure 1). The maps further distinguish between four classes of claims: cognitive (i.e., claims about cognitive/psychological functions; colored in cyan), substrate-general (i.e., claims about a generally defined "physical substrate", without assuming that it must be a brain; green), brain-specific (i.e., claims about the brain and its properties; pink), and extrapolation (conjectures about the distribution of consciousness in domains beyond that of healthy adult humans; see Baetu, 2024; Bayne et al., 2024; Negro & Mudrik, 2025; yellow).

The classification of each level on the map depends on the nature of the theory. For theories centered mostly around cognitive/psychological claims (GNWT, RPT, and PRM), the third level (L3) captures cognitive predictions. In this case, L4 focuses on theoretical claims about the neural implementation of the cognitive claims (L3), while the fifth and outermost level is devoted to experimentally testable neuroscientific predictions. On the other hand, if the central regions of the map are dominated by substrate-general claims (IIT), the third level is composed of brain-specific claims, the fourth of neuroscientific predictions, and the fifth of cognitive predictions.

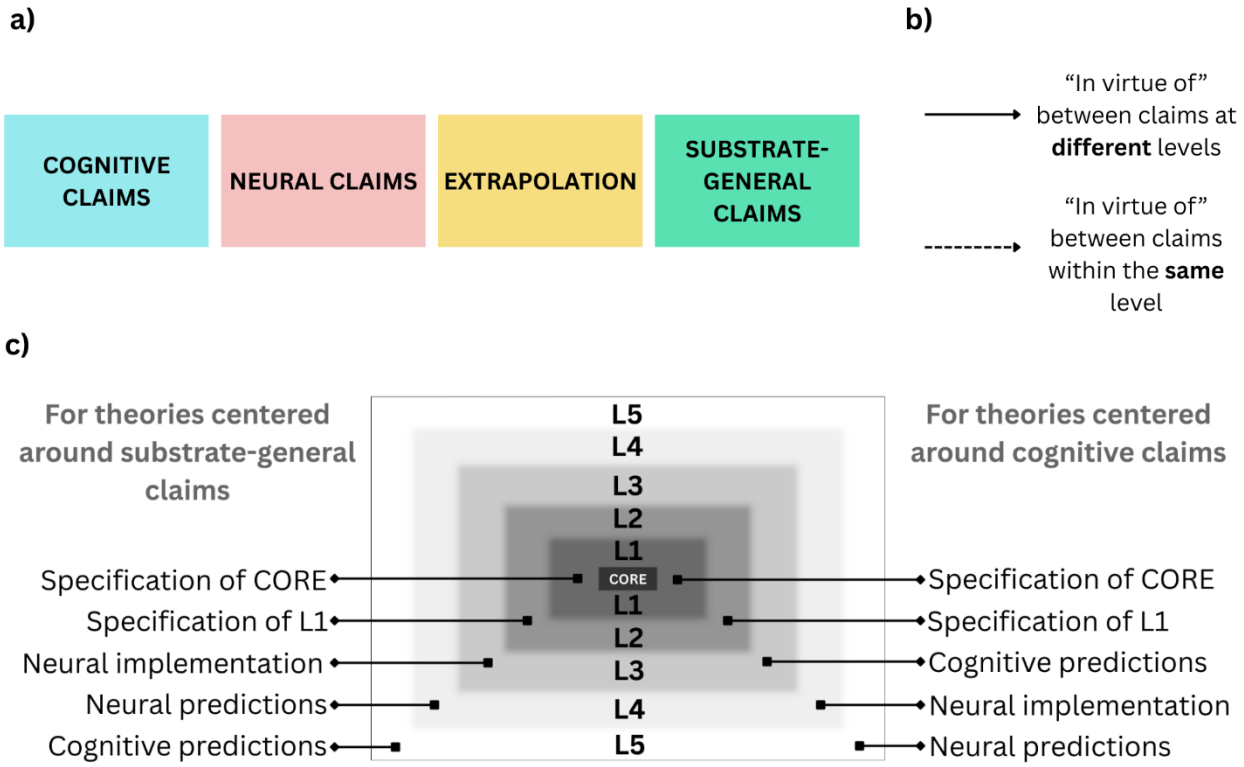


Figure 1. *Annotation of prediction maps.* **(a)** *different types of claims present in the maps.* **(b)** *meaning of the connectors (arrows) between claims.* **(c)** *meaning of levels. Left: layout for theories centered on substrate-general claims. Right: layout for cognitive/functional theories.*

We represented the logical relations between core, auxiliary claims, and predictions through a generic "in virtue of" relation, conveying the logical (but not necessarily deductive) dependence of a claim on other claims. This relation can occur both between claims at two different levels (represented by a solid line arrow) or between claims belonging to the same level (represented by a dashed line arrow).

To minimize the subjective aspect of map construction and to make sure our reconstruction was not biased or mistaken, we contacted the theory-proponents and asked them to provide feedback on the maps. The final versions of the prediction maps, reported here, are the result of several rounds of revisions and informative exchanges with theory-proponents.

Centrality measures

To quantitatively evaluate the structural role of each claim within the prediction maps, we applied graph-theoretical centrality measures within a network analysis framework (Borgatti &

Everett, 2006; Freeman, 1978). Centrality is one of the main indicators of influential nodes based on the topological properties that drive network dynamics (Bloch et al., 2023; Ghalmane et al., 2019; Gurfinkel & Rikvold, 2021). Such analysis assigns a score to each claim based on different aspects of its structural position and potential influence in the network (Rodrigues, 2019).

In the context of prediction maps, centrality measures allow us to address the following questions (Borgatti & Everett, 2006): How widely does a given claim influence other parts of the theory? How quickly can its influence propagate? Does it serve as a critical connector between separate parts of the map? Is it supported by highly influential claims? Correspondingly, we computed four complementary centrality measures, which are applied at the level of single claims in the prediction maps (as opposed to the whole network level): Local Reaching Centrality and Harmonic Centrality, which assess the breadth and immediacy of a claim's downstream influence; Betweenness Centrality, which highlights its function as a connector between theoretical components; and PageRank, providing an estimate of the importance of each node based on the quality of its upstream support. Below we define and explain each measure (Figure 2).

Local Reaching Centrality (LRC) and Harmonic Centrality (HC)

LRC captures the reachability of a given node by calculating the total number of nodes it can reach (Mones et al., 2012, Equation 1; figure 3a).

$$\mathrm{LRC(i)} = \frac{Number\ of\ nodes\ reached\ by\ i}{n-1} \tag{1}$$

This measure highlights nodes that can reach a large portion of the map, thereby identifying claims that serve as starting points for broad logical implications. Because of its intuitive interpretation, LRC is particularly useful for comparing how far different claims extend within the structure of the theory (Mones et al., 2012). Yet, it only offers a partial view: it does not reflect how far away those nodes lie and how efficiently they connect to the rest of the network.

HC goes beyond LRC by taking into account how far away the nodes are from each other: instead of treating all reachable nodes equally, HC weights each node by the inverse of its shortest distance. Specifically, it sums the reciprocals of the shortest path lengths from each node i to every node j that can be reached (see Equation 2; Figure 3b). This means that nodes reached in fewer steps contribute more strongly (e.g., 1/1 for a direct neighbor) than those reached via longer paths (e.g., 1/3 if three steps away) (Rochat, 2009). To account for network size, this sum is normalized by the maximum possible number of nodes that could be reached.

$$\text{Harmonic Centrality}(i) = \frac{1}{n-1} \sum_{j \in R(i)} \frac{1}{d_{ij}} \quad (2)$$

where R(i) denotes the set of nodes that can be reached from node i, d_{ij} denotes the shortest path length from node i to node j, and n denotes the total number of nodes in the network.

In the context of prediction maps, HC highlights claims whose influence can propagate quickly through the network. Unlike LRC, which simply counts how many nodes a claim can reach, HC also considers how far away those nodes are, giving higher weight to claims that reach others in fewer steps. It captures both the breadth of influence and the efficiency of reach, making it a valuable complement to LRC. Accordingly, we report them together here.

Betweenness Centrality (BC)

BC measures the importance of a node in connecting other nodes in the network. Namely, it indicates to what extent a certain claim serves as a "logical bridge". It quantifies how often a node is present on the shortest paths between other pairs of nodes (Freeman, 1977, Equation 3; figure 3c). Therefore, BC captures the role of a node as a crucial link in the transmission of information.

$$BC(i) = \frac{2}{(n-1)(n-2)} \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}} \quad (3)$$

where σ_{st} denotes the number of shortest paths from node *s* to node *t*, $\sigma_{st}(i)$ denotes the number of shortest paths from node *s* to node *t* passing through node i, *and* n denotes the total number of nodes in the network.

When applied to prediction maps, BC indicates which claims serve as crucial logical interconnectors between different parts of the theory. Thus, providing empirical evidence against nodes with high BC amounts to disabling crucial logical connections in the theory's structure.

PageRank (PR)

PageRank (PR) is an algorithm used by Google Search to rank web pages in their search engine results. PR recursively evaluates each node based on the number and importance of nodes linking to it, reflecting the idea that a claim is important if it is supported by other influential claims, thus capturing a form of authority within the theoretical structure rather than sheer connectivity (Page et al., 1999, Equation 4; figure 3d). PR is formalized in the following way:

$$PR(i) = \frac{1 - d}{\mathrm{n}} + d * \sum_{j \in Predecessors(i)} \frac{PR(j)}{L(j)} \quad (4)$$

where $Predecessors(i)$ denotes the nodes that point to i, $PR(j)$ denotes the PageRank of node j (importance of the claim linking to i), $L(j)$ denotes the number of outgoing edges from node j (how many claims j justifies), *n* denotes the total number of nodes in the network, and *d* is the damping factor (i.e. the probability that a user will move to the linked web page. Accordingly, 1-d is the probability of choosing a random web page). Here, *d* is set to 0.85 following (Brin & Page, 1998).

PR first assigns equal weights to all nodes and then repeatedly updates these weights using the above equation until the values stabilize at an equilibrium, which reflects the final importance of each claim.

In prediction maps, PR highlights claims that are repeatedly justified by other well-supported claims, thereby identifying downstream nodes that play a central role in supporting the overall theoretical framework. This is important because challenging predictions with high PR reverberates directly to other highly influential upstream claims.

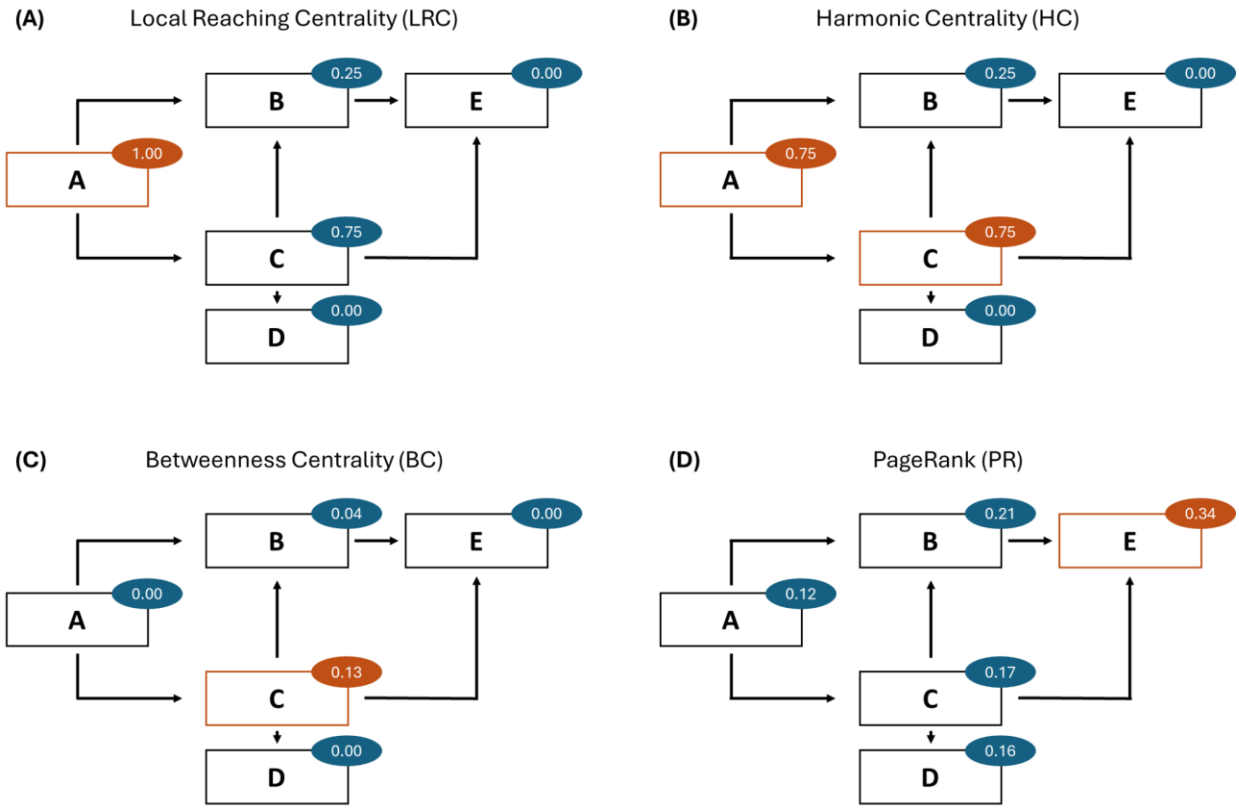


Figure 2. *An example for a simple directed network used to illustrate different centrality measures. The same network is shown in all panels; in each panel, the node with the highest score is highlighted in orange.* **(A) Local Reaching Centrality (LRC).** *node A has the highest LRC since it can reach all other nodes in the network. Node E, on the other hand, does not reach any other node, and accordingly has an LRC score of zero.* **(B) Harmonic Centrality (HC).** *Nodes A and C have the highest HC; Although A can reach all other nodes, and C can reach all nodes except A, their HC values are the same because HC accounts not only for reachability but also for path length. In particular, C reaches several nodes via shorter paths than A. For example, both A and C can reach node D; however, A reaches D in two steps (A → C → D), whereas C reaches D in one step (C → D). Because HC weights shorter paths more strongly than longer ones, C's more efficient access to downstream nodes compensates for its slightly smaller reachable set.* **(C) Betweenness Centrality (BC).** *Node C has the highest BC because it lies on multiple shortest paths between other node pairs. For example, both the shortest path from A to E and the one from A to D pass through C (A → C → E; A → C → D). As a result, C acts as a critical connector between upstream node A and downstream nodes D and E.* **(D) PageRank (PR).** *Node E attains the highest PR since it is*

supported by, or based on, both B and C, and those nodes are themselves well supported. That is, B is supported by both A and C, and thus has a relatively high rank, and because B itself only supports node E - this entire rank is transferred to E. In addition, C also links directly to E (C → E), thereby further increasing E's rank.

Transparency and openness

All analysis code is available and added as supplemental resource to the manuscript. This study was not preregistered.

Prediction Maps

In this section, we present the prediction maps for the four theories, and analyze the centrality of each of their claims (for the full centrality analysis, see Supplementary S1). For convenience, claims are labelled alphanumerically, based on the level they belong to (i.e., A for the core, B for claims in L1, etc.), while the numbering progresses clockwise from the top-left claim in each level. Thus, the labels should not be mistaken to convey logical connections, which are represented by the arrows (so, for example, C4 can lead to C3, in which case an arrow will go from the former to the latter). Each claim is supported by a numbered reference, according to the reference lists reported in Table 1. In addition, though we attempted to expose auxiliary assumptions, we focused on those relevant to the theory, and did not attempt to exhaust all possible assumptions. For example, transitioning between cognitive or substrate-general claims and neural-specific claims requires the hidden assumption that the human brain is the most likely organ to sustain the relevant cognitive function (but see Rouleau & Levin, 2023). This assumption is based on decades of neuroscientific research that cannot be represented in prediction maps, and is accordingly left out.

GNWT

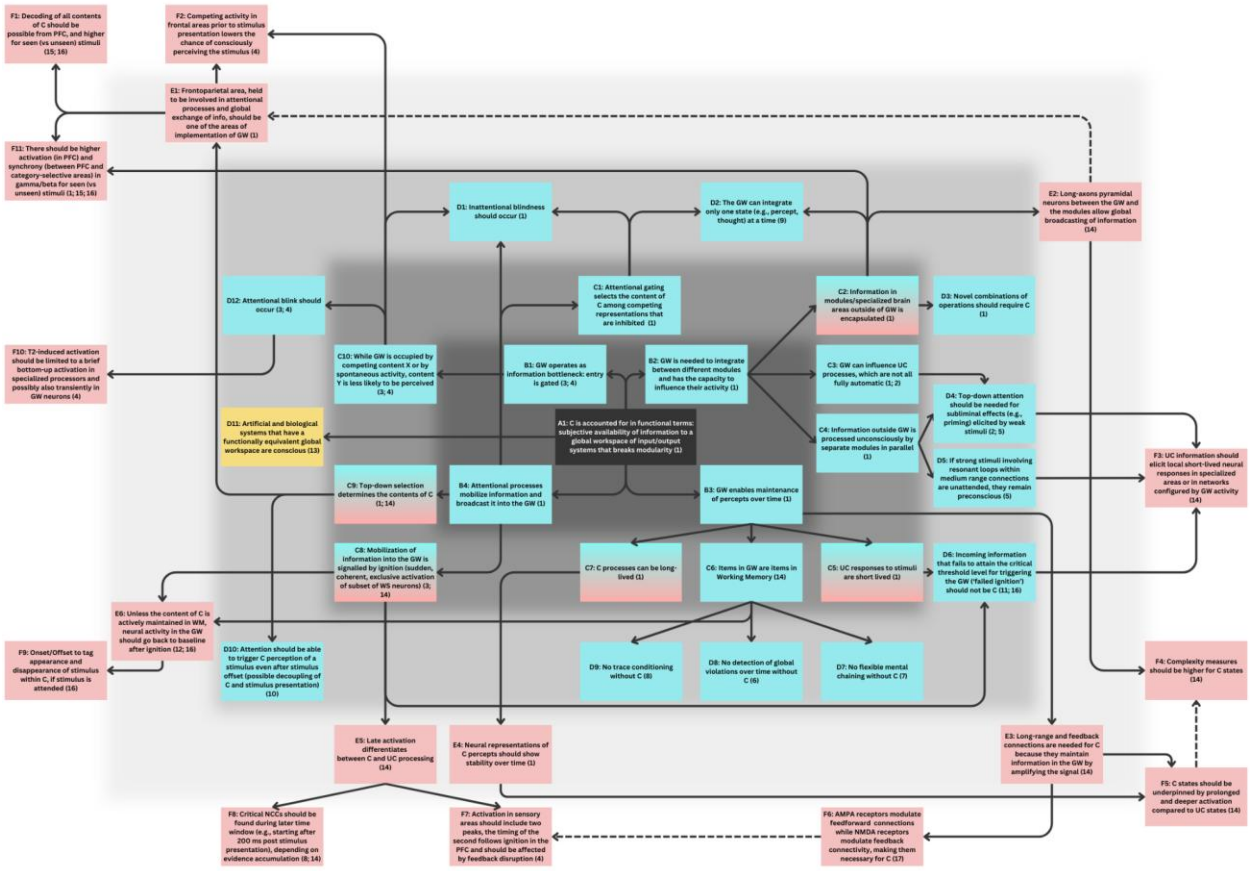


Figure 3. *GNWT Prediction map. Abbreviations: C (consciousness/conscious); GW (global workspace); NCC (neural correlates of consciousness); PFC (prefrontal cortex); UC (unconsciousness; unconscious); WM (working memory); WS (workspace). See Table 1 for references.*

The black rectangle at the center of the map (Figure 3), denoting the core claim of GNWT (**A1**), highlights its functionalist standpoint by suggesting that the function of consciousness is making information available by broadcasting it into a global workspace (GW) that breaks modularity. Consequently, most of the theory's claims in the layers closer to the core (L1 and L2) are cognitive, specifying the processes through which the GW operates: **B1** states that the GW functions as an information bottleneck; **B2** claims that the GW is needed to integrate between different modules; **B3** states that the GW enables maintenance of percepts over time; **B4** specifies the role of attentional processes – they mobilize information and broadcast it into the GW.

Two of the claims in L1 (B2 and B3) lead to many other cognitive claims down the line, suggesting that they are important "hubs" for the theory. The **B2** cluster includes claims about unconscious processes and how they are affected by the GW. Within this cluster, claims **C4** ("Information outside GW is processed unconsciously by separate modules in parallel") and **C3** ("GW can influence unconscious processes, which are not all fully automatic"), lead to two cognitive predictions (**D4**, **D5**). These two predictions capture important aspects of how the theory considers unconscious processes: **D4** asserts that subliminal effects, contrary to purely automatic processes, require attention; **D5** stresses that if stimuli involving resonant loops within medium range connections are unattended, they remain preconscious.

The cluster stemming from **B3**, instead, focuses on the maintenance of information over time. The fecundity of this cluster is evidenced by the number and types of claims **B3** generates. Indeed, **B3** leads to cognitive claims **C5** ("UC responses to stimuli are short-lived"), **C6** ("Items in GW are items in Working Memory"), and **C7** ("Conscious processes can be long-lived"), as well as to the neural implementation claim **E3** ("Long-range and feedback connections are needed for consciousness because they maintain information in the GW by amplifying the signal"). Among these claims, the most consequential one is **C6**, which brings about the neural implementation claim **E6** and three cognitive predictions (**D7**, **D8**, **D9**) about specific instances where certain capacities (trace conditioning, detection of global violations over time and flexible mental chaining) can only take place if the information is consciously maintained.

LRC and HC analysis

The centrality analysis shows that not all claims connected to the core have a similar importance, as within L1, there is high variability between the different claims. **B4**, which describes the role of attention in the theory, connects the core to more downstream claims than any other claim in the map, despite not belonging to a thematically unified cluster ($B4_{LRC}$= 0.465; $B4_{HC}$= 0.260). In contrast, claim **B1** (concerning GW operating as an information bottleneck) is not connected to many nodes in the map ($B1_{LRC}$= 0.116; $B1_{HC}$= 0.066), and its HC score is lower than that of the ensuing **C10**, which specifies how the bottleneck actually operates ($C10_{LRC}$= 0.093; $C10_{HC}$= 0.081). This suggests that **B1**'s impact on the overall theory is mediated by **C10**. Together, the LRC and HC measures indicate that for GNWT, claims about attention, information maintenance

and integration reach more claims than the claim that the GW acts as a bottleneck, despite the prominence of this claim in discussions surrounding GNWT (Dehaene et al., 2003; Dehaene & Changeux, 2005).

In L2, four more claims (**C2**, **C8**, **C6**, **C9**) can be identified as having relatively high levels of LRC and HC (proportional to the general levels within the map). Out of these four, claim **C2** has the highest reach within its level ($C2_{LRC}$= 0.186; $C2_{HC}$= 0.130), suggesting that the claim that information in modules outside the GW is encapsulated is highly consequential. Moreover, **E2** has LRC and HC levels ($E2_{LRC}$= 0.116; $E2_{HC}$= 0.081) that resemble those of claim **B1**, despite **B1** being close to the core and **E2** being a brain-specific claim (L4). This highlights the centrality of this claim, concerning long-axon pyramidal neurons allowing global broadcasting of information. It also exemplifies that consequential claims are not constrained to the levels that are closest to the theory's core: implementation claims can be highly influential if they have multiple empirical consequences.

BC analysis

The BC analysis indicates that in GNWT, the claims receiving a high BC value are mostly the ones highlighted above (**B2-B4**, **C10**, **C2** and also **C8** and **E1**), with **B4** reaching the highest value (0.008). This shows that in GNWT, claims that have far-reaching consequences also act as connectors between different areas of the map. Claim **E1**, which describes the frontoparietal network as a likely implementation site of the GW, is particularly interesting: because of its high BC score (0.007), it can be seen as the crucial connector between key theoretical *cognitive* concepts of GNWT and its *neural* predictions.

PR analysis

Finally, the PR analysis shows that the highest level of PR (0.084) is found for neural prediction **F3**, which concerns unconscious information eliciting local short-lived neural responses. This claim is the only one derived from three other important claims (**D4-D6**). Accordingly, falsifying this prediction would have a strong impact on the map. Other predictions that score high in this metric are the neural predictions **F4** (PR =0.054) and **F5** (PR =0.041). Falsifying either one of

these (especially **F4**, as it would reverberate to **F5**) would challenge many important constructs of the theory.

IIT

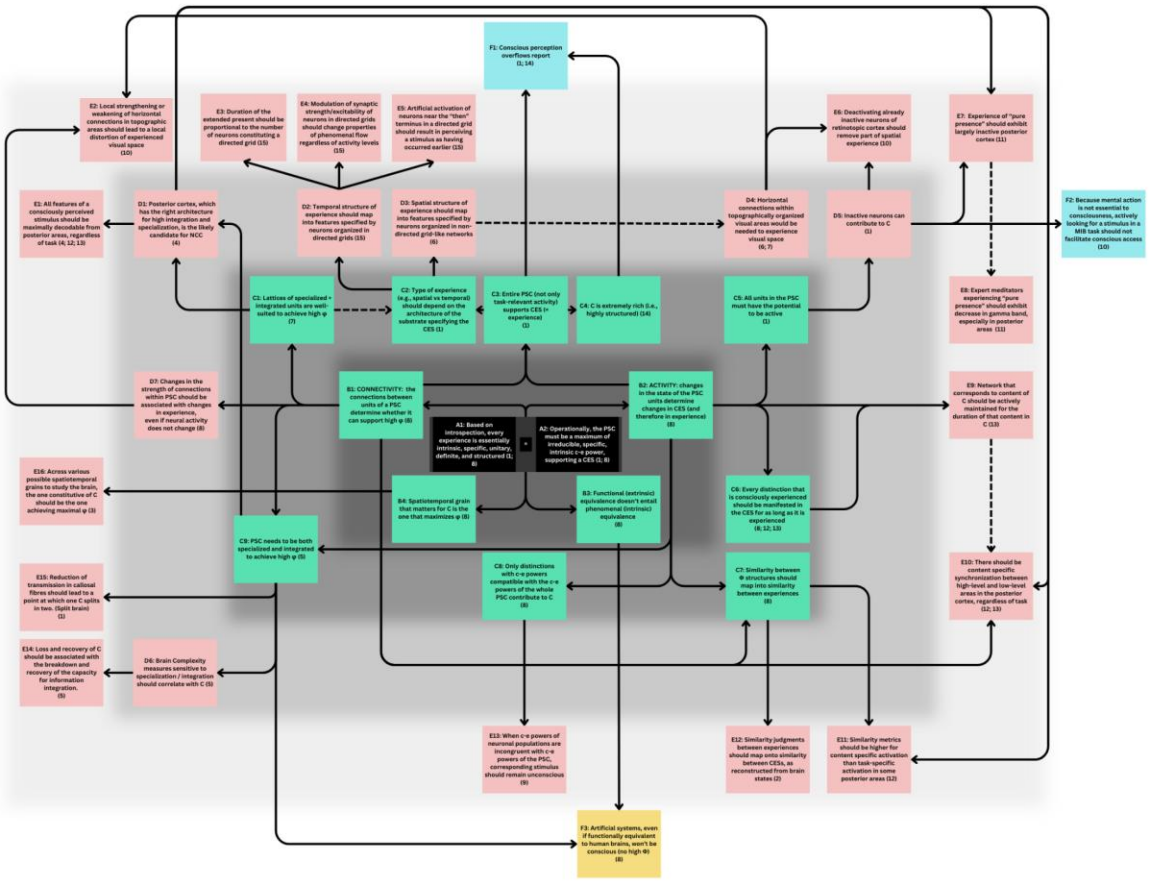


Figure 4. *IIT Prediction map. Abbreviations: C (consciousness); C-E (cause-effect); CES (cause-effect structure); NCC (neural correlates of consciousness); PSC (physical substrate of consciousness). See Table 1 for references.*

IIT's core (Figure 4) is constituted by two sets of claims: the axioms (**A1**, representing the essential properties of consciousness) and the postulates (**A2**, representing the corresponding physical operationalization). The equals sign between the two boxes in the core conveys the explanatory identity IIT posits between phenomenology and the cause-effect structure (CES), such that the properties of this structure explain all properties of consciousness without any further additional ingredient. The next level, L1, focuses on key properties that the physical substrate of consciousness must satisfy to support consciousness. Because these claims pertain to the physical substrate supporting conscious experience, but not necessarily the brain, we coded

them as "substrate-general" claims (in green). Note that in this map, we follow IIT's usage of the Greek letter "phi" as a measure of consciousness (Albantakis et al., 2023), distinguishing between φ (small phi), when referring to the physical substrate, and Φ (big phi), when referring to the cause-effect structure specified by the physical substrate (i.e., PSC: physical substrate of consciousness).

The transition from the principles of IIT, specified in CORE, L1, and L2, to neural terms requires regarding neurons as units and synapses as connections. To illustrate, the substrate-general claim **C5** about *units* contributing to consciousness if they have the potential to be active, translates into a claim about inactive *neurons* contributing to consciousness (**D5**). In turn, prediction **D5** warrants both neural-specific predictions that can be tested experimentally (**E6** and **E7**) and, with the help of auxiliaries not explicitly represented here, the cognitive prediction **F2**.

The map also reveals groups of related claims within IIT. For example, a cluster of claims targets the question of *whether* a system is conscious or not. This cluster is enabled by claim **C9** ("PSC needs to be both specialized and integrated to achieve high φ") and includes an extrapolation claim to other populations (e.g., artificial systems; **F3**), as well as claims about neurophysiological edge cases (**E15**) and clinical settings (**E14**, **D6**).

A different cluster, emerging from claim **C2,** concerns the relationship between the type of experience and the architecture of the corresponding substrate. This cluster characterizes the types of contents in a conscious experience, thus targeting the question of *what* an experience feels like (e.g., its temporal structure (**D2**), its duration (**E3**), its spatial character (**D3**), etc.).

LRC and HC analysis

Claims **B1** (about the state of units in the PSC) and **B2** (about the connectivity between units) play a particularly prominent role in connecting the core to many claims in subsequent levels, whereas claims **B3** and **B4**, despite being located near the core, are not heavily connected to the rest of the map: the LRC and HC analysis reveals that **B2** is the most productive node ($B2_{LRC}$= 0.821; $B2_{HC}$= 0.430, compared to the second most productive node $B1_{LRC}$= 0.692; $B1_{HC}$= 0.362) in the IIT map, excluding the core, which by definition reaches all nodes of the network. This suggests that for IIT, claims about the *state* of the units of a physical substrate are more far-reaching and direct than claims about the *connectivity* between units of a physical substrate. On

the contrary, **B3** and **B4** both score quite low in LRC and HC ($B3_{LRC}$= 0.026; $B3_{HC}$= 0.026; $B4_{LRC}$= 0.026; $B4_{HC}$= 0.026), indicating that they are not particularly productive claims in terms of empirical consequences. Notably, three claims in L2 score high in LRC and HC, despite their more peripheral location: **C1** ("Lattices of specialized and integrated units are well-suited to achieve high φ"; $C1_{LRC}$= 0.385; $C1_{HC}$= 0.184), **C3** ("C3: the entire physical substrate of consciousness (not only task-relevant activity) supports the cause-effect structure (= experience)"; $C3_{LRC}$= 0.282; $C3_{HC}$= 0.150), and **C9** ("the physical substrate of consciousness needs to be both specialized and integrated to achieve high φ"; $C9_{LRC}$= 0.256; $C9_{HC}$= 0.175).

BC analysis

Among the most influential claims acting as crucial bridges between different parts of the map (**C2**, **B2**, and **C3**, with the first scoring BC= 0.024 and the others BC= 0.020), **C2** ("Type of experience (e.g., spatial vs temporal) should depend on the architecture of the substrate specifying the cause-effect structure") deserves particular attention, as it indicates that the relationship between the cause-effect structure and the architecture of the physical substrate of consciousness is crucial to bridge the theoretical and the empirically testable aspects of IIT.

Pairing this BC analysis with the individuation of clusters identified above shows that the two clusters are not equally important for IIT: **C2**, enabling the first cluster, scores 0.024 in BC, while **C9**, enabling the second cluster, scores 0.013. This means that empirical investigations over what an experience feels like are more informative to IIT than investigations over whether consciousness is present or not.

PR analysis

This analysis identifies high-stakes predictions of IIT, as they derive from other important claims in the network. The neural-specific predictions with the highest PR value is **E10** ("There should be content-specific synchronization between high-level and low-level areas in the posterior cortex, regardless of the task"), with a value of PR= 0.049. This means that falsifying this prediction would have a strong impact on the map insofar as it relies on crucial theoretical constructs (e.g., the connectivity of the physical substrate of consciousness and the posterior cortex being the most likely NCC candidate). Other predictions that score high in this metric are the neural predictions **E2** ("Local strengthening or weakening of horizontal connections in

topographic areas should lead to a local distortion of experienced visual space"; $E2_{PR}$= 0.045), **E6** ("Deactivating already inactive neurons of retinotopic cortex should remove part of spatial experience"; $E6_{PR}$= 0.038) and **E8** ("Expert meditators experiencing "pure presence" should exhibit decrease in gamma band, especially in posterior areas"; $E8_{PR}$= 0.038), as well as the brain implementation claim **D4** ("Horizontal connections within topographically organized visual areas would be needed to experience visual space"; $D4_{PR}$= 0.037) and the cognitive prediction **F1** ("Conscious perception overflows report"; $F1_{PR}$= 0.037). Each of these predictions stem from paths formed by influential claims: **F1** relies heavily on **C3**, while both **E6** and **E8** rely on IIT's bold claim **D5**, namely that inactive neurons can contribute to consciousness. Again, refuting these predictions would reverberate on crucial nodes of IIT's map.

PRM

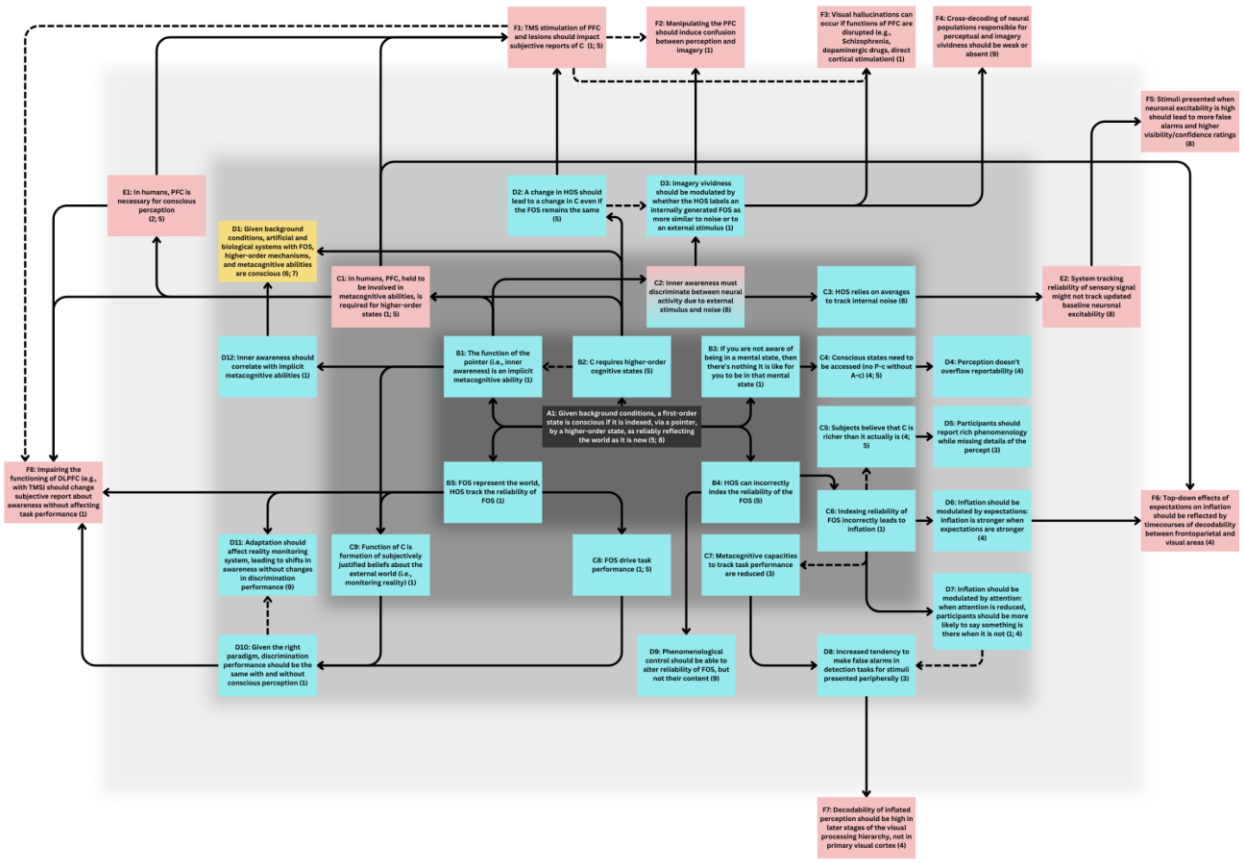


Figure 5. *PRM Prediction map. Abbreviations: A-c (access consciousness); C (consciousness); FOS (first-order state); HOS (higher-order state); P-c (phenomenal consciousness); PFC (prefrontal cortex); TMS (transcranial magnetic stimulation). See Table 1 for references.*

At the core of various Higher-Order-Thought theories is the idea that first-order states are not conscious unless they are referred to in a specific manner by a higher-order state (Brown et al., 2019). The specific relationship between the first- and higher-order states that constitutes conscious awareness is different for each version of HOT. According to PRM, on which we focus here (for other HOTs, see supplementary S3), the higher-order state functions as "reality monitoring", and it indexes first-order states according to whether they reliably reflect the world (Figure 5). Thus, the core claim of PRM, **A1**, is that a first-order state is conscious if it is indexed, via a pointer, by a higher-order state, as reliably reflecting the world as it is now. The next level, L1, comprises five cognitive claims (**B1-B5**) elaborating on the core and specifying the functional relationship between first-order and higher-order states.

L1 serves as a gateway for addressing key questions about consciousness (e.g., phenomenal vs access consciousness, see Block, 1995; the richness of conscious experience, see Kouider et al., 2010; the function of consciousness, see Shepherd & Bayne, 2026), that are then fleshed out by claims in L2.

The map reveals two major clusters of claims. The first stems from the cognitive claims **B1** and **B2**, and is mostly concerned with the neural implementation of the theory in the human brain. This cluster has multiple connections to other paths in the map. For example, one of the neuroscientific predictions belonging to this cluster is **F8** ("Impairing the functioning of the dorsolateral prefrontal cortex should change subjective report about awareness without affecting task performance"), which is also connected to claims **B5** ("First-order states represent the world, higher-order states track the reliability of first-order states") and **D10** ("Given the right paradigm, discrimination performance should be the same with and without conscious perception"), which do not belong to the same thematic cluster. Another example is **C2** ("Inner awareness must discriminate between neural activity due to external stimulus and noise"), which branches out to three claims concerning signal-noise discrimination.

The second cluster, stemming from **B4** ("Higher-order states can incorrectly index the reliability of first-order states"), focuses on richness and inflation. It is mediated by the claim (**C6**) that indexing the reliability of first order states incorrectly leads to inflation, and includes the claims **C5** ("Subjects believe that consciousness is richer than it actually is") and **C7** ("Metacognitive capacities to track task performance are reduced"). The path leads to the cognitive predictions

D5 ("Participants should report rich phenomenology while missing details of the percept"), **D6** ("Inflation should be modulated by expectations: inflation is stronger when expectations are stronger"), **D7** ("Inflation should be modulated by attention: when attention is reduced, participants should be more likely to say something is there when it is not"), **D8** ("Increased tendency to make false alarms in detection tasks for stimuli presented peripherally"), **D9** ("Phenomenological control should be able to alter reliability of first-order states, but not their content"), and the two neural predictions **F6** ("Top-down effects of expectations on inflation should be reflected by time courses of decodability between frontoparietal and visual areas") and **F7** ("Decodability of inflated perception should be high in later stages of the visual processing hierarchy, not in primary visual cortex").

LRC and HC analysis

Besides the core, **B2** is the claim with the highest values in both measures ($B2_{LRC}$= 0.556, $B2_{HC}$= 0.288), followed closely by **B1** ($B1_{LRC}$= 0.500, $B1_{HC}$= 0.275). Their high value stems from the aforementioned inter-connected cluster of claims they generate, as well as from another, smaller path about the discrimination between signal and noise, starting at **C2** and ending in the neural prediction **F5** ("Stimuli presented when neuronal excitability is high should lead to more false alarms and higher visibility/confidence ratings").

Other claims in L1 lead to much fewer claims. Notably, **B3**, which captures the transitivity principle (Rosenthal, 2000), leads to the fewest ($B3_{LRC}$= 0.056, $B3_{HC}$= 0.042) - only two, concerning phenomenal vs. access consciousness. **B4** leads to more claims ($B4_{LRC}$= 0.278, $B4_{HC}$= 0.146), though only via the mediation of **C6** (also reflected by its higher HC value, $C6_{HC}$= 0.162), leading to the relatively encapsulated path about richness and inflation.

Nodes located further away from the core have notably high LRC and HC scores. These are **C1** ($C1_{LRC}$= 0.167, $C1_{HC}$= 0.134), **C2** ($C2_{LRC}$= 0.194, $C2_{HC}$= 0.120), **C6** ($C6_{LRC}$= 0.222, $C6_{Hc}$ =0.162), and the cognitive prediction **D2** ("A change in the higher-order state should lead to a change in consciousness even if the first-order state remains the same"; $D2_{LRC}$= 0.167, $D2_{HC}$= 0.111).

BC analysis

Many claims reach similar high values of BC. The highest is achieved by **B1** (0.014), which directly leads to the big, interconnected cluster about neural implementation. Because **B1** stems from **B2**, it acts as a bridge connecting **B2** to all of the claims in this cluster, further demonstrating the importance of connecting such a foundational theoretical claim ("Consciousness requires higher-order cognitive states") to claims about neural implementation. The other important cluster-connector is **C6**, which reaches a BC value of 0.012 and connects **B4** to a group of cognitive claims and predictions. This analysis suggests that investigations over the neural implementation of PRM could be more telling than studies on subjective inflation.

PR analysis

Three of the four nodes identified as high-stakes predictions in the PR analysis are neural. The highest value belongs to **F8** ($F8_{PR}$= 0.063), supported by two paths - first, the neural-implementation path, starting at **B1** and passing crucially through **C1**; second, the branch stemming from **B5**, specifying the distinct functional role of first- and higher-order states, involving the cognitive prediction **D10** ("Given the right paradigm, discrimination performance should be the same with and without conscious perception").

The second and third highest PR scores belong to **F3** ($F3_{PR}$= 0.052; "Visual hallucinations can occur if functions of the prefrontal cortex are disrupted, e.g., Schizophrenia, dopaminergic drugs, direct cortical stimulation") and **F7** ("Decodability of inflated perception should be high in later stages of the visual processing hierarchy, not in primary visual cortex", $F7_{PR}$= 0.051). If refuted, these predictions would indirectly affect important constructs of the theory: **F3** would influence the distinction between perception and imagery, supposedly tracked by the reality monitoring system; **F7**, on the other hand, would put pressure on prediction **D8**, which depends on how inflation works.

Finally, **D10** ($D10_{PR}$= 0.046) is also central to the theory. This claim has a bearing on **C9** (the theory's view on the function of consciousness) and by implication also on **B5**, which defines the relationship between higher- and first-order states for this particular version of HOT. This makes it a very specific claim, and therefore important when comparing PRM against other HOTs.

RPT

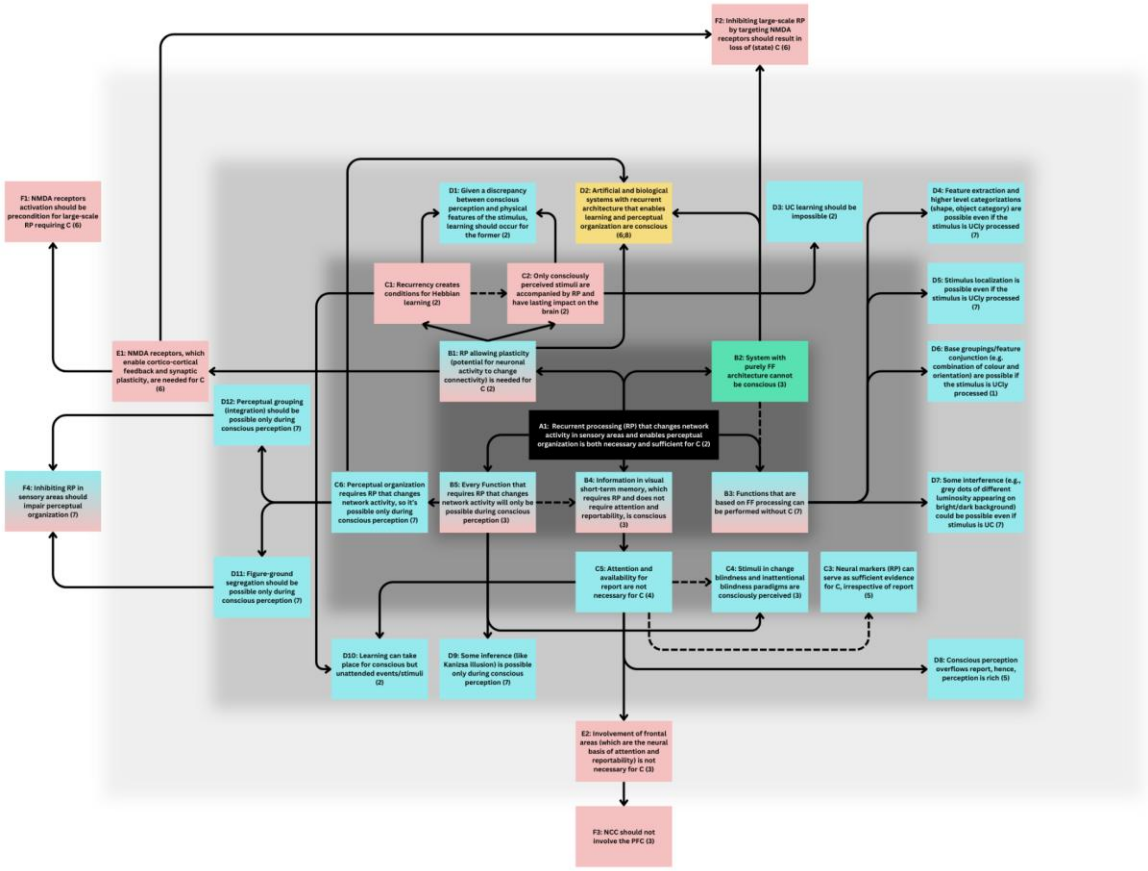


Figure 6: *RPT Prediction map. Abbreviations: C (consciousness); FF (feedforward); NCC (neural correlates of consciousness); PFC (prefrontal cortex); RP (recurrent processing); UC (unconscious). See Table 1 for references.*

The core idea of RPT (Figure 6) describes recurrent processing that changes network activity in sensory areas and enables perceptual organization as both necessary and sufficient for consciousness (**A1**). The next level, L1, focuses on the neural and architectural requirements derived from this core and is thus characterized mainly by hybrid claims (i.e., including both cognitive and neural aspects). Two of these claims further emphasize that plasticity and feedback processes are essential for conscious perception, rather than mere neural activation (**B1**: "RP allowing plasticity (potential for neuronal activity to change connectivity) is needed for consciousness", **B2**: "System with purely feedforward architecture cannot be conscious"). The other claims in L1 detail cognitive *functions* that depend on recurrent processing.

The map reveals several clusters of claims within RPT. One such cluster, enabled by claim **B3** ("Functions that are based on feedforward processing can be performed without consciousness"), concerns functional distinctions between conscious (recurrent processing-dependent) and unconscious (feedforward-based) processes. For instance, RPT posits that high-level categorization (**D4**), stimulus localization (**D5**), feature conjunction (**D6**), and interference (**D7**) are all possible without consciousness. The map shows that these claims are all derived from one cognitive principle, captured by **B3**.

Another important conceptual cluster concerns the involvement of attentional processes in consciousness. This is particularly important since it helps differentiate RPT from cognitive theories like GNWT. The cluster (composed of **C3**, **C4**, **C5**, **D8**, **D9**, **D10**, **E2**, **F3**) emerges from **B4** ("Information in visual short-term memory, which requires RP and does not require attention and reportability, is conscious"), and emphasizes the dissociation between consciousness and attention/reportability. This dissociation leads to both cognitive and neural assertions.

LRC and HC analysis

This analysis highlights both **B1** and **B5** as central hubs, connecting the core to many downstream predictions. This further demonstrates the ability of the maps and the centrality measures to capture the essence of the theories, as RPT was previously described as building upon two aspects: recurrency and plasticity/learning (Lamme, 2018).

B5 ("Every Function that requires recurrent processing that changes network activity will only be possible during conscious perception") is the most consequential node ($B5_{LRC}$ =0.483; $B5_{HC}$ = 0.273), predicting the cognitive capacities that can be performed only via recurrent processing (paralleling the first cluster described above).

B1 ($B1_{LRC}$ =0.310; $B1_{HC}$ =0.218) anchors RPT's learning and plasticity principles. **B1** leads directly to **C1** ("Recurrency creates conditions for Hebbian learning") and **C2** ("Only consciously perceived stimuli are accompanied by recurrent processing and have lasting impact on the brain"). **B1** also connects to the extrapolation (**D2**: "Artificial and biological systems with recurrent architecture that enables learning and perceptual organization are conscious") and molecular mechanisms (**E1**: "NMDA receptors, which enable cortico-cortical feedback and

synaptic plasticity, are needed for consciousness"), making it pivotal for RPT's cross-domain generalization.

A particularly interesting case is **C5** ("Attention and availability for report are not necessary for consciousness"), which, despite its peripheral position in L2, shows high LRC ($C5_{LRC}$ =0.207) and HC ($C5_{HC}$ =0.190). This reflects how RPT's attentional independence claim serves as an important connector between cognitive and neural clusters, influencing both experimental predictions (**D8, D10, F3**) and theoretical formulations (**E2, C3, C4**).

BC analysis

C5 has the highest betweenness score in the map ($C5_{BC}$ =0.019), confirming its connector role between otherwise weakly linked clusters. It is the only path linking the *attention–reportability* subnetwork (**C3, D8, E2, F3**) with upstream theoretical claims. This highlights **C5**'s empirical importance: falsifying it would disconnect parts of the map and force a major revision of RPT's architecture.

Interestingly, the other cluster identified above stems from claim **B3**, which reaches a lower BC score (0.010). This means that empirically targeting C5 could be more meaningful for RPT than targeting **B3**, because the "attention cluster" is more relevant than the "unconscious functions cluster".

PR analysis

Notably, downstream predictions within RPT typically depend on multiple upstream claims, meaning that refuting any single prediction can simultaneously challenge several related claims. Specifically, **F4** ("Inhibiting recurrent processing in sensory areas should impair perceptual organization") has the highest PR ($F4_{PR}$ =0.069) since multiple mid-level claims converge exclusively on this prediction; **F4** relies on both **D11** ("Figure-ground segregation should be possible only during conscious perception") and **D12** ("Perceptual grouping (integration) should be possible only during conscious perception"), offering experimentally testable claims regarding the disruption of consciousness through targeted interventions. Thus, refuting **F4** would cast doubt on these claims, which will in turn affect the more central claim **C6**. Two other high PR nodes, **C5** ($C5_{PR}$ =0.046) and **F3** ("NCC should not involve the prefrontal cortex"; $F3_{PR}$

=0.045), highlight the influential status of the claims about irrelevance of attentional/reportability mechanisms (**C5**) and of the prefrontal cortex (**F3**) for consciousness.

Discussion

This work introduces prediction maps as means to expose the logical relationship between core claims of scientific theories and their empirical predictions. These maps visualize the informativeness of different claims of the theory (e.g., which claims yield more downstream claims, whether each claim is supported by a single or multiple "parent" claims, etc.). Thus, experimentalists can use prediction maps to design direct tests of a theory, theorists might employ them to revise their theories in light of contradictory evidence, and scholars in general can use them to compare and assess the structure of competing scientific theories.

A key advantage afforded by the maps is the ability to quantify the centrality of each claim using graph-theory measures (Borgatti & Everett, 2006), thereby understanding its importance for theory evaluation. Notably, these measures go beyond assessing the closeness of a given claim to the core and instead quantify its connectivity profile. Given the different meanings of each centrality measure, researchers should focus on the measure of interest based on their experimental and theoretical goals: LRC and HC score how far-reaching each claim is (for HC, this is weighted by the "distance" between claims), BC measures the extent to which a claim works as a crucial connector, while PR indicates which nodes are the most important given the importance of their parents.

Centrality measures are informative insofar as the network to which they are applied accurately represents the corresponding theory. In this sense, they are sensitive to subjective modeler decisions (Parker & Winsberg, 2018), and their interpretation depends on how a theory is represented in the first place. Accordingly, centrality-based inferences should be taken with a grain of salt. In our case, we tried to minimize subjectivity by having the maps constructed by multiple modelers, and by discussing the maps with theory leaders. We suggest that future applications of our approach take similar steps to mitigate this concern.

We hold that applying prediction maps to scientific theories can advance theory evaluation across many disciplines and fields. It might be especially suited for disciplines like psychology and neuroscience, given the qualitative nature of central theoretical constructs and the lack of

formal models regulating the relationship between these constructs (Meehl, 1967; Oberauer & Lewandowsky, 2019). Specifically, here we focused on theories of consciousness, which are still "works in progress" (Mudrik et al., 2025), underspecified and empirically under-constrained (Cleeremans et al., 2025; Fazekas et al., 2024; Kirkeby-Hinrup et al., 2025; Negro et al., 2025). Below, we accordingly use the insights derived from this process to demonstrate how prediction maps can accelerate scientific progress, by advancing theory understanding and evaluation.

First, visual inspection of the four maps we generated reveals the deep theoretical and conceptual divergence between the theories: GNWT and PRM have a functional/cognitive core, surrounded by purely cognitive claims. In contrast, RPT's core idea is informed by both cognitive and neural constructs, and accordingly four out of the five claims in L1 are hybrid claims, combining cognitive and neural constructs. Finally, IIT stands out, since its core is based on its phenomenological axioms and the explanatory identity posited between properties of phenomenology and the properties of a cause-effect structure. Thus, its L1 claims are substrate-general.

Second, the maps also allow us to more readily detect convergence points between the theories, while also assessing the differential value of such points for each theory. For example, both RPT and GNWT posit that NMDA receptors play a role in generating consciousness (for RPT, see **E1** and **F1**; for GNWT, see **F6**). With centrality measures, we can now assess the relative importance of such convergence points for each theory: for this example, RPT's **E1** ("NMDA receptors, which enable cortico-cortical feedback and synaptic plasticity, are needed for C") scores relatively high in both LRC (0.241; 90th percentile: 0.248; median: 0.017) and HC (0.149; 90th percentile: 0.193; median: 0.017). On the contrary, GNWT's **F6** ("AMPA receptors modulate feedforward connections while NMDA receptors modulate feedback connectivity, making them necessary for C") has a medium value across all centrality measures (LRC =0.023; HC =0.023; BC =0.001; PR =0.019; all values are around the median for each measure - see Table 17 in Supplementary S1). This demonstrates that although both GNWT and RPT agree that NMDA receptors are necessary for consciousness, this claim plays a much bigger role in RPT than in GNWT.

Third, despite being merely descriptive (rather than evaluative), prediction maps can be used for further evaluation, and specifically for detecting weak links in the current formulations of the

theories. For example, for GNWT it is unclear why the construct of ignition leads to the prediction that the disappearance of a stimulus should lead to an ignition-like response, creating a gap between **E6** and **F9**. For IIT, there seems to be a wide gap between **B2** and **C7**: additional claims are needed to more explicitly justify why similarity between Phi-structures and experiences should derive from the fact that changes in the state of the PSC determine changes in experience. For PRM, it is unclear why decodability of inflated perception should be high in later stages of the visual hierarchy, rendering the transition between **D8** and **F7** obscure. Finally, for RPT, there seems to be a weak link between **E1** and **F1**, since it is not specified what counts as "large-scale" recurrent processing - how "large" is large enough?

Moreover, the centrality analysis (see Supplementary S1), and especially the PR scores, provides strong bases for evaluating existing attempts to test theories. In the case of theories of consciousness, several adversarial collaborations have been trying to evaluate such theories (Corcoran et al., 2025; Fleming et al., 2025; Melloni et al., 2021; Olcese et al., 2024; Potgieter, 2024). The first published work emanating from these efforts (Cogitate et al., 2025), testing predictions of IIT and GNWT, has been criticized for not targeting predictions that are equally informative to the tested theories (IIT-Concerned et al., 2025). Specifically, it was argued that "the tests are problematic and not meaningful because they do not address the core claims of IIT" (IIT-Concerned et al., 2025, p. 689). However, our analysis shows that IIT's **E10**, which corresponds to the third prediction tested in the Cogitate experiment, actually reaches the highest PR score in the IIT map (0.049), suggesting it is the most important prediction to target if one wants to meaningfully test IIT (interestingly, it was not confirmed by the data). The other two IIT predictions tested by the Cogitate experiment are **E9**, which has a medium-high PR value (0.031; 90th percentile: 0.037; median: 0.022), and **E1**, with a medium PR value (0.020). In comparison, the GNWT predictions tested by the Cogitate experiment seem to be less central to the theory. **F9**, the second prediction in the first Cogitate experiment, reaches a medium-high PR value (0.030; 90th percentile: 0.033; median: 0.018), while the first (**F1**) and third (**F11**) have a medium PR value ($F1_{PR}$= 0.020; $F11_{PR}$= 0.023). This suggests that if anything, the predictions tested by Cogitate in the first experiment were somewhat more central to IIT than to GNWT. This seems to be the case also for Experiment 2 (see Mudrik et al., 2024), which tested the same predictions for GNWT (**F1, F9** and **F11**), and an additional prediction for IIT, **E11** ("Similarity metrics should be higher for content-specific activation than task-specific activation in some

posterior areas"), achieving a medium-high PR score of 0.028. Irrespective of the debate on the Cogitate results, this example showcases that prediction maps and centrality analyses can inform debates surrounding theory testing, paving the way towards more systematic conceptual and quantifiable assessments (See Supplementary S2 for a similar analysis based on predictions tested by other adversarial collaborations).

On the other hand, the informativeness of theoretical predictions for theory evaluation is not only defined by centrality; it is also influenced by other properties (Syrjänen, 2024) like novelty (Douglas & Magnus, 2013; Worrall, 1989), boldness (Mayo, 1991; Popper, 1959), and precision (Hitchcock & Sober, 2004; but see Elliott-Graves, 2020). Taking a wider perspective, theory evaluation is not confined to the predictive power of a theory, and involves other theoretical virtues like parsimony and coherence with background knowledge (Keas, 2018; Longino, 1979). And at an even higher level, scientific progress can be attained pre-theoretically, even without theory assessment and theory comparison (Chang, 2004). Thus, the current approach, focused on the centrality of predictions and their relevance for scientific theories, is only one way to achieve scientific progress when facing unresolved questions and unexplained phenomena (Figure 7).

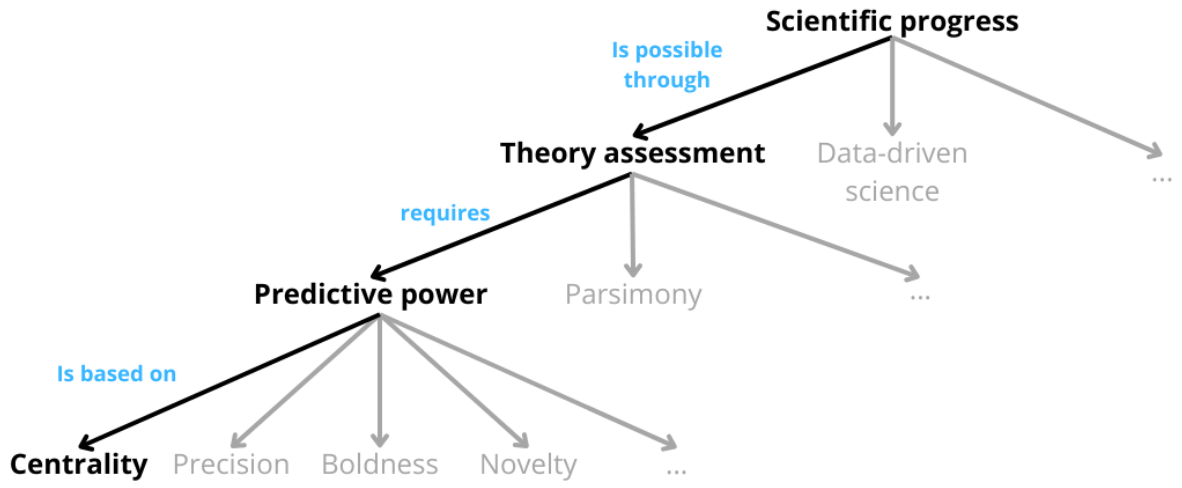


Figure 7. *A possible path towards scientific progress. Progress is partly driven by theory assessment, which requires establishing, among other things (e.g., explanatory power, parsimony, coherence with background knowledge, etc.), the predictive power of scientific theories, which is a multidimensional concept comprising the importance of the centrality, novelty, boldness, and precision of predictions for theory testing. Here, we focus on the centrality dimension.*

Finally, we acknowledge the limitations of the prediction maps approach, and suggest future directions that can compensate for those limitations. First, they are tuned to challenging theories rather than confirming them (as the entire approach is inspired by Lakatosian falsificationism; see Lakatos, 1976). Bayesian confirmation theory can be used to complement our approach in that respect (Grim et al., 2022; Henderson et al., 2010; Corcoran et al., 2023; Kirkeby-Hinrup, 2024). Second, the maps represent one point in time. Instead, theories can be represented diachronically, with different prediction maps representing the different versions of the theory at different time points (e.g., IIT has been updated several times: Tononi, 2004, 2012; Oizumi et al., 2014; Albantakis et al., 2023). This could enable a Lakatos-inspired assessment of whether the research program has progressed over time (Negro, 2024). Third, our prediction maps do not differentiate between prediction and accommodation; while the former is formulated without considering the results, the latter is designed to fit the already obtained data (Hitchcock & Sober, 2004). In our maps, we call "prediction" any empirical statement logically derived from upstream claims. Future maps might want to include the distinction between predictions and accommodations. Fourth, these maps do not include all auxiliary assumptions that further specify the relationship between a given hypothesis and the claims from which it is derived. Fifth, our graph-theoretic analysis focused exclusively on within-network measures (i.e., by asking how influential a claim is for *this* network). Whole-network measures could be also exploited to compare network structures, thus enabling deeper theory-comparison. Taken together, these future directions highlight how our approach, applied here to theories of consciousness, can be extended to exploring other questions and theoretical dimensions. We hope prediction maps will thus become a key tool for theory testing and evaluation across different domains and disciplines.

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Supplemental materials

1. Centrality tables for each theory
2. Centrality analysis for predictions of IIT and PRM tested in various ACs
3. Additional theories (table with references and maps)

Supplementary S1: Centrality tables for each theory

For each theory, we report here the full list of claims with their corresponding centrality scores, according to the four measures used in this work (see Main text, "Centrality measures").

GNWT

Claim	LRC
Core: Consciousness is accounted for in functional terms: subjective availability of information to a global workspace of input/output systems that breaks modularity (1)	1.000
B4: Attentional processes mobilize information and broadcast it into the GW (1)	0.465
B3: GW enables maintenance of percepts over time (1)	0.372
B2: GW is needed to integrate between different modules and has the capacity to influence their activity (1)	0.326
C2: Information in modules/specialized brain areas outside of GW is encapsulated (1)	0.186
C8: Mobilization of information into the GW is signalled by ignition (sudden, coherent, exclusive activation of subset of WS neurons) (3; 14)	0.163
E2: Long-axon pyramidal neurons between the GW and the modules allow global broadcasting of information (14)	0.116
C6: Items in GW are items in Working Memory (14)	0.116
C9: Top-down selection determines the contents of C (1; 14)	0.116
B1: GW operates as information bottleneck: entry is gated (3; 4)	0.116
C10: While GW is occupied by competing content X or by spontaneous activity, content Y is less likely to be perceived (3; 4)	0.093
E3: Long-range and feedback connections are needed for C because they maintain information in the GW by amplifying the signal (14)	0.093
E1: Frontoparietal area, held to be involved in attentional processes and global exchange of info, should be one of the areas of implementation of GW (1)	0.070

C4: Information outside GW is processed unconsciously by separate modules in parallel (1)	0.070
C7: C processes can be long-lived (1)	0.070
E4: Neural representations of C percepts should show stability over time (1)	0.047
E5: Late activation differentiates between C and UC processing (14)	0.047
C3: GW can influence UC processes, which are not all fully automatic (1; 2)	0.047
C5: UC responses to stimuli are short-lived (1)	0.047
C1: Attentional gating selects the content of C among competing representations that are inhibited (1)	0.047
F5: C states should be underpinned by prolonged and deeper activation compared to UC states (14)	0.023
D4: Top-down attention should be needed for subliminal effects (e.g., priming) elicited by weak stimuli (2; 5)	0.023
D6: Incoming information that fails to attain the critical threshold level for triggering the GW ('failed ignition') should not be C (11; 15)	0.023
D12: Attentional blink should occur (3; 4)	0.023
E6: Unless the content of C is actively maintained in WM, neural activity in the GW should go back to baseline after ignition (12; 15)	0.023
D5: If strong stimuli involving resonant loops within medium range connections are unattended, they remain preconscious (5)	0.023
F6: AMPA receptors modulate feedforward connections while NMDA receptors modulate feedback connectivity, making them necessary for C (16)	0.023
F3: UC information should elicit local short-lived neural responses in specialized areas or in networks configured by GW activity (14)	0.000
F4: Complexity measures should be higher for C states (14)	0.000
F7: Activation in sensory areas should include two peaks, the timing of the second follows ignition in the PFC and should be affected by feedback disruption (4)	0.000
F10: T2-induced activation should be limited to a brief bottom-up activation in specialized processors and possibly also transiently in GW neurons (4)	0.000
F9: Onset/Offset to tag appearance and disappearance of stimulus within C, if stimulus is attended (15)	0.000
D1: Inattentional blindness should occur (1)	0.000
F2: Competing activity in frontal areas prior to stimulus presentation lowers the chance of consciously perceiving the stimulus (4)	0.000
F11: There should be higher activation (in PFC) and synchrony (between PFC and category-selective areas) in gamma/beta for seen (vs unseen) stimuli (1; 15; 16)	0.000
D2: The GW can integrate only one state (e.g., percept, thought) at a time (9)	0.000

F1: Decoding of all contents of C should be possible from PFC, and higher for seen (vs unseen) stimuli (15)	0.000
F8: Critical NCCs should be found during later time window (e.g., starting after 200 ms post stimulus presentation), depending on evidence accumulation (8; 14)	0.000
D10: Attention should be able to trigger C perception of a stimulus even after stimulus offset (possible decoupling of C and stimulus presentation) (10)	0.000
D3: Novel combinations of operations should require C (1)	0.000
D9: No trace conditioning without C (8)	0.000
D8: No detection of global violations over time without C (6)	0.000
D7: No flexible mental chaining without C (7)	0.000
D11: Artificial and biological systems that have a functionally equivalent global workspace are conscious (13)	0.000

Table S1.1. *GNWT Local Reach Centrality. Claims in light blue are the ones tested by the COGITATE adversarial collaboration. Abbreviations and references are reported in the main text.*

Claim	HC
Core: Consciousness is accounted for in functional terms: subjective availability of information to a global workspace of input/output systems that breaks modularity (1)	0.440
B4: Attentional processes mobilize information and broadcast it into the GW (1)	0.260
B3: GW enables maintenance of percepts over time (1)	0.220
B2: GW is needed to integrate between different modules and has the capacity to influence their activity (1)	0.170
C2: Information in modules/specialized brain areas outside of GW is encapsulated (1)	0.130
C8: Mobilization of information into the GW is signalled by ignition (sudden, coherent, exclusive activation of subset of WS neurons) (3; 14)	0.116
C6: Items in GW are items in Working Memory (14)	0.105
E2: Long-axon pyramidal neurons between the GW and the modules allow global broadcasting of information (14)	0.081
C9: Top-down selection determines the contents of C (1; 14)	0.081
C10: While GW is occupied by competing content X or by spontaneous activity, content Y is less likely to be perceived (3; 4)	0.081
E3: Long-range and feedback connections are needed for C because they maintain information in the GW by amplifying the signal (14)	0.070
E1: Frontoparietal area, held to be involved in attentional processes and global exchange of info, should be one of the areas of implementation of GW (1)	0.070
B1: GW operates as information bottleneck: entry is gated (3; 4)	0.066

C4: Information outside GW is processed unconsciously by separate modules in parallel (1)	0.058
E5: Late activation differentiates between C and UC processing (14)	0.047
C1: Attentional gating selects the content of C among competing representations that are inhibited (1)	0.047
C7: C processes can be long-lived (1)	0.043
E4: Neural representations of C percepts should show stability over time (1)	0.035
C3: GW can influence UC processes, which are not all fully automatic (1; 2)	0.035
C5: UC responses to stimuli are short-lived (1)	0.035
F5: C states should be underpinned by prolonged and deeper activation compared to UC states (14)	0.023
D4: Top-down attention should be needed for subliminal effects (e.g., priming) elicited by weak stimuli (2; 5)	0.023
D6: Incoming information that fails to attain the critical threshold level for triggering the GW ('failed ignition') should not be C (11; 15)	0.023
D12: Attentional blink should occur (3; 4)	0.023
E6: Unless the content of C is actively maintained in WM, neural activity in the GW should go back to baseline after ignition (12; 15)	0.023
D5: If strong stimuli involving resonant loops within medium range connections are unattended, they remain preconscious (5)	0.023
F6: AMPA receptors modulate feedforward connections while NMDA receptors modulate feedback connectivity, making them necessary for C (16)	0.023
F3: UC information should elicit local short-lived neural responses in specialized areas or in networks configured by GW activity (14)	0.000
F4: Complexity measures should be higher for C states (14)	0.000
F7: Activation in sensory areas should include two peaks, the timing of the second follows ignition in the PFC and should be affected by feedback disruption (4)	0.000
F10: T2-induced activation should be limited to a brief bottom-up activation in specialized processors and possibly also transiently in GW neurons (4)	0.000
F9: Onset/Offset to tag appearance and disappearance of stimulus within C, if stimulus is attended (15)	0.000
D1: Inattentional blindness should occur (1)	0.000
F2: Competing activity in frontal areas prior to stimulus presentation lowers the chance of consciously perceiving the stimulus (4)	0.000
F11: There should be higher activation (in PFC) and synchrony (between PFC and category-selective areas) in gamma/beta for seen (vs unseen) stimuli (1; 15; 16)	0.000
D2: The GW can integrate only one state (e.g., percept, thought) at a time (9)	0.000

F1: Decoding of all contents of C should be possible from PFC, and higher for seen (vs unseen) stimuli (15)	0.000
F8: Critical NCCs should be found during later time window (e.g., starting after 200 ms post stimulus presentation), depending on evidence accumulation (8; 14)	0.000
D10: Attention should be able to trigger C perception of a stimulus even after stimulus offset (possible decoupling of C and stimulus presentation) (10)	0.000
D3: Novel combinations of operations should require C (1)	0.000
D9: No trace conditioning without C (8)	0.000
D8: No detection of global violations over time without C (6)	0.000
D7: No flexible mental chaining without C (7)	0.000
D11: Artificial and biological systems that have a functionally equivalent global workspace are conscious (13)	0.000

Table S1.2. *GNWT Harmonic Centrality.*

Claim	BC
B4: Attentional processes mobilize information and broadcast it into the GW (1)	0.008
E1: Frontoparietal area, held to be involved in attentional processes and global exchange of info, should be one of the areas of implementation of GW (1)	0.007
B3: GW enables maintenance of percepts over time (1)	0.007
C2: Information in modules/specialized brain areas outside of GW is encapsulated (1)	0.007
C8: Mobilization of information into the GW is signalled by ignition (sudden, coherent, exclusive activation of subset of WS neurons) (3; 14)	0.006
C10: While GW is occupied by competing content X or by spontaneous activity, content Y is less likely to be perceived (3; 4)	0.006
B2: GW is needed to integrate between different modules and has the capacity to influence their activity (1)	0.005
C6: Items in GW are items in Working Memory (14)	0.005
E2: Long-axon pyramidal neurons between the GW and the modules allow global broadcasting of information (14)	0.005
C9: Top-down selection determines the contents of C (1; 14)	0.004
E3: Long-range and feedback connections are needed for C because they maintain information in the GW by amplifying the signal (14)	0.004
E5: Late activation differentiates between C and UC processing (14)	0.003
E6: Unless the content of C is actively maintained in WM, neural activity in the GW should go back to baseline after ignition (12; 15)	0.003

F5: C states should be underpinned by prolonged and deeper activation compared to UC states (14)	0.002
D6: Incoming information that fails to attain the critical threshold level for triggering the GW ('failed ignition') should not be C (11; 15)	0.002
C4: Information outside GW is processed unconsciously by separate modules in parallel (1)	0.002
D12: Attentional blink should occur (3; 4)	0.002
C5: UC responses to stimuli are short-lived (1)	0.001
D4: Top-down attention should be needed for subliminal effects (e.g., priming) elicited by weak stimuli (2; 5)	0.001
F6: AMPA receptors modulate feedforward connections while NMDA receptors modulate feedback connectivity, making them necessary for C (16)	0.001
B1: GW operates as information bottleneck: entry is gated (3; 4)	0.001
C7: C processes can be long-lived (1)	0.001
E4: Neural representations of C percepts should show stability over time (1)	0.001
C3: GW can influence UC processes, which are not all fully automatic (1; 2)	0.001
C1: Attentional gating selects the content of C among competing representations that are inhibited (1)	0.001
D5: If strong stimuli involving resonant loops within medium range connections are unattended, they remain preconscious (5)	0.001
Core: Consciousness is accounted for in functional terms: subjective availability of information to a global workspace of input/output systems that breaks modularity (1)	0.000
F3: UC information should elicit local short-lived neural responses in specialized areas or in networks configured by GW activity (14)	0.000
F4: Complexity measures should be higher for C states (14)	0.000
F7: Activation in sensory areas should include two peaks, the timing of the second follows ignition in the PFC and should be affected by feedback disruption (4)	0.000
F10: T2-induced activation should be limited to a brief bottom-up activation in specialized processors and possibly also transiently in GW neurons (4)	0.000
F9: Onset/Offset to tag appearance and disappearance of stimulus within C, if stimulus is attended (15)	0.000
D1: Inattentional blindness should occur (1)	0.000
F2: Competing activity in frontal areas prior to stimulus presentation lowers the chance of consciously perceiving the stimulus (4)	0.000
F11: There should be higher activation (in PFC) and synchrony (between PFC and category-selective areas) in gamma/beta for seen (vs unseen) stimuli (1; 15; 16)	0.000

D2: The GW can integrate only one state (e.g., percept, thought) at a time (9)	0.000
F1: Decoding of all contents of C should be possible from PFC, and higher for seen (vs unseen) stimuli (15)	0.000
F8: Critical NCCs should be found during later time window (e.g., starting after 200 ms post stimulus presentation), depending on evidence accumulation (8; 14)	0.000
D10: Attention should be able to trigger C perception of a stimulus even after stimulus offset (possible decoupling of C and stimulus presentation) (10)	0.000
D3: Novel combinations of operations should require C (1)	0.000
D9: No trace conditioning without C (8)	0.000
D8: No detection of global violations over time without C (6)	0.000
D7: No flexible mental chaining without C (7)	0.000
D11: Artificial and biological systems that have a functionally equivalent global workspace are conscious (13)	0.000

Table S1.3. *GNWT Betweenness Centrality.*

Claim	PR
F3: UC information should elicit local short-lived neural responses in specialized areas or in networks configured by GW activity (14)	0.084
F4: Complexity measures should be higher for C states (14)	0.054
F5: C states should be underpinned by prolonged and deeper activation compared to UC states (14)	0.041
F7: Activation in sensory areas should include two peaks, the timing of the second follows ignition in the PFC and should be affected by feedback disruption (4)	0.036
D4: Top-down attention should be needed for subliminal effects (e.g., priming) elicited by weak stimuli (2; 5)	0.034
D6: Incoming information that fails to attain the critical threshold level for triggering the GW ('failed ignition') should not be C (11; 15)	0.030
F10: T2-induced activation should be limited to a brief bottom-up activation in specialized processors and possibly also transiently in GW neurons (4)	0.030
F9: Onset/Offset to tag appearance and disappearance of stimulus within C, if stimulus is attended (15)	0.030
D1: Inattentional blindness should occur (1)	0.029
F2: Competing activity in frontal areas prior to stimulus presentation lowers the chance of consciously perceiving the stimulus (4)	0.028
C10: While GW is occupied by competing content X or by spontaneous activity, content Y is less likely to be perceived (3; 4)	0.028
E4: Neural representations of C percepts should show stability over time (1)	0.026

E1: Frontoparietal area, held to be involved in attentional processes and global exchange of info, should be one of the areas of implementation of GW (1)	0.026
F11: There should be higher activation (in PFC) and synchrony (between PFC and category-selective areas) in gamma/beta for seen (vs unseen) stimuli (1; 15; 16)	0.023
D2: The GW can integrate only one state (e.g., percept, thought) at a time (9)	0.023
D12: Attentional blink should occur (3; 4)	0.020
E6: Unless the content of C is actively maintained in WM, neural activity in the GW should go back to baseline after ignition (12; 15)	0.020
F1: Decoding of all contents of C should be possible from PFC, and higher for seen (vs unseen) stimuli (15)	0.020
F8: Critical NCCs should be found during later time window (e.g., starting after 200 ms post stimulus presentation), depending on evidence accumulation (8; 14)	0.020
D5: If strong stimuli involving resonant loops within medium range connections are unattended, they remain preconscious (5)	0.020
F6: AMPA receptors modulate feedforward connections while NMDA receptors modulate feedback connectivity, making them necessary for C (16)	0.019
D10: Attention should be able to trigger C perception of a stimulus even after stimulus offset (possible decoupling of C and stimulus presentation) (10)	0.019
E5: Late activation differentiates between C and UC processing (14)	0.017
C2: Information in modules/specialized brain areas outside of GW is encapsulated (1)	0.017
C4: Information outside GW is processed unconsciously by separate modules in parallel (1)	0.017
C3: GW can influence UC processes, which are not all fully automatic (1; 2)	0.017
E2: Long-axon pyramidal neurons between the GW and the modules allow global broadcasting of information (14)	0.016
D3: Novel combinations of operations should require C (1)	0.016
D9: No trace conditioning without C (8)	0.016
D8: No detection of global violations over time without C (6)	0.016
D7: No flexible mental chaining without C (7)	0.016
C6: Items in GW are items in Working Memory (14)	0.016
E3: Long-range and feedback connections are needed for C because they maintain information in the GW by amplifying the signal (14)	0.016
C5: UC responses to stimuli are short-lived (1)	0.016
C7: C processes can be long-lived (1)	0.016
C8: Mobilization of information into the GW is signalled by ignition (sudden, coherent, exclusive activation of subset of WS neurons) (3; 14)	0.015

C9: Top-down selection determines the contents of C (1; 14)	0.015
C1: Attentional gating selects the content of C among competing representations that are inhibited (1)	0.015
B4: Attentional processes mobilize information and broadcast it into the GW (1)	0.015
B3: GW enables maintenance of percepts over time (1)	0.015
B2: GW is needed to integrate between different modules and has the capacity to influence their activity (1)	0.015
B1: GW operates as information bottleneck: entry is gated (3; 4)	0.015
D11: Artificial and biological systems that have a functionally equivalent global workspace are conscious (13)	0.015
Core: Consciousness is accounted for in functional terms: subjective availability of information to a global workspace of input/output systems that breaks modularity (1)	0.013

Table S1.4. *GNWT PageRank.*

IIT

Claim	LRC
Core: Based on introspection, every experience is essentially intrinsic, specific, unitary, definite, and structured = Operationally, the PSC must be a maximum of irreducible, specific, intrinsic c-e power, supporting a CES (1; 8)	1.000
B2: ACTIVITY: changes in the state of the PSC units determine changes in CES (and therefore in experience) (8)	0.821
B1: CONNECTIVITY: the connections between units of a PSC determine whether it can support high phi (8)	0.692
C1: Lattices of specialized + integrated units are well-suited to achieve high phi (7)	0.385
C3: Entire PSC (not only task-relevant activity) supports CES (= experience) (1)	0.282
C9: PSC needs to be both specialized and integrated to achieve high phi (5)	0.256
C2: Type of experience (e.g., spatial vs temporal) should depend on the architecture of the substrate specifying the CES (1)	0.205
D1: Posterior cortex, which has the right architecture for high integration and specialization, is the likely candidate for NCC (4)	0.128
C5: All units in the PSC must have the potential to be active (1)	0.128
D5: Inactive neurons can contribute to consciousness (1)	0.103
D2: Temporal structure of experience should map into features specified by neurons organized in directed grids (15)	0.077

D3: Spatial structure of experience should map into features specified by neurons organized in non-directed grid-like networks (6)	0.077
D4: Horizontal connections within topographically organized visual areas would be needed to experience visual space (6; 7)	0.051
C7: Similarity between PHI structures should map into similarity between experiences (8)	0.051
C6: Every distinction that is consciously experienced should be manifested in the CES for as long as it is experienced (8; 12; 13)	0.051
E9: Network that corresponds to content of C should be actively maintained for the duration of that content in C (13)	0.026
E7: Experience of 'pure presence' should exhibit largely inactive posterior cortex (11)	0.026
C4: C is extremely rich (i.e., highly structured) (14)	0.026
D6: Brain Complexity measures sensitive to specialization / integration should correlate with consciousness (5)	0.026
B4: Spatiotemporal grain that matters for consciousness is the one that maximizes phi (8)	0.026
B3: Functional (extrinsic) equivalence doesn't entail phenomenal (intrinsic) equivalence (8)	0.026
D7: Changes in the strength of connections within PSC should be associated with changes in experience, even if neural activity does not change (8)	0.026
C8: Only distinctions with c-e powers compatible with the c-e powers of the whole PSC contribute to C (8)	0.026
E10: There should be content-specific synchronization between high-level and low-level areas in the posterior cortex, regardless of the task (12; 13)	0.000
E2: Local strengthening or weakening of horizontal connections in topographic areas should lead to a local distortion of experienced visual space (10)	0.000
E8: Expert meditators experiencing 'pure presence' should exhibit decrease in gamma band, especially in posterior areas (11)	0.000
E6: Deactivating already inactive neurons of retinotopic cortex should remove part of spatial experience (10)	0.000
F1: Conscious perception overflows report (1; 14)	0.000
F3: Artificial systems, even if functionally equivalent to human brains, won't be conscious (no high PHI) (8)	0.000
E14: Loss and recovery of consciousness should be associated with the breakdown and recovery of the capacity for information integration. (5)	0.000
E16: Across various possible spatiotemporal grains to study the brain, the one constitutive of consciousness should be the one achieving maximal phi (3)	0.000

E13: When c-e powers of neuronal populations are incongruent with c-e powers of the PSC, corresponding stimulus should remain unconscious (9)	0.000
E11: Similarity metrics should be higher for content-specific activation than task-specific activation in some posterior areas (12)	0.000
E12: Similarity judgments between experiences should map onto similarity between CESs, as reconstructed from brain states (2)	0.000
F2: Because mental action is not essential to consciousness, in a MIB task actively looking for a stimulus should not facilitate conscious access (10)	0.000
E5: Artificial activation of neurons near the then terminus in a directed grid should result in perceiving a stimulus as having occurred earlier (15)	0.000
E4: Modulation of synaptic strength/excitability of neurons in directed grids should change properties of phenomenal flow regardless of activity levels (15)	0.000
E3: Duration of the extended present should be proportional to the number of neurons constituting a directed grid (15)	0.000
E1: All features of a consciously perceived stimulus should be maximally decodable from posterior areas, regardless of task (4; 12; 13)	0.000
E15: Reduction of transmission in callosal fibres should lead to a point at which one consciousness splits in two. (Split brain) (1)	0.000

Table S1.5. *IIT Local Reach Centrality. Claims in light blue are the ones tested by the COGITATE adversarial collaboration. Claims in lilac are tested by INTREPID. Abbreviations and references are reported in the main text.*

Claim	HC
B2: ACTIVITY: changes in the state of the PSC units determine changes in CES (and therefore in experience) (8)	0.430
Core: Based on introspection, every experience is essentially intrinsic, specific, unitary, definite, and structured = Operationally, the PSC must be a maximum of irreducible, specific, intrinsic c-e power, supporting a CES (1; 8)	0.421
B1: CONNECTIVITY: the connections between units of a PSC determine whether it can support high phi (8)	0.362
C1: Lattices of specialized + integrated units are well-suited to achieve high phi (7)	0.184
C9: PSC needs to be both specialized and integrated to achieve high phi (5)	0.175
C3: Entire PSC (not only task-relevant activity) supports CES (= experience) (1)	0.150
C2: Type of experience (e.g., spatial vs temporal) should depend on the architecture of the substrate specifying the CES (1)	0.120
D1: Posterior cortex, which has the right architecture for high integration and specialization, is the likely candidate for NCC (4)	0.115
D5: Inactive neurons can contribute to consciousness (1)	0.090

D2: Temporal structure of experience should map into features specified by neurons organized in directed grids (15)	0.077
C5: All units in the PSC must have the potential to be active (1)	0.073
D3: Spatial structure of experience should map into features specified by neurons organized in non-directed grid-like networks (6)	0.051
D4: Horizontal connections within topographically organized visual areas would be needed to experience visual space (6; 7)	0.051
C7: Similarity between PHI structures should map into similarity between experiences (8)	0.051
C6: Every distinction that is consciously experienced should be manifested in the CES for as long as it is experienced (8; 12; 13)	0.038
E9: Network that corresponds to content of C should be actively maintained for the duration of that content in C (13)	0.026
E7: Experience of 'pure presence' should exhibit largely inactive posterior cortex (11)	0.026
C4: C is extremely rich (i.e., highly structured) (14)	0.026
D6: Brain Complexity measures sensitive to specialization / integration should correlate with consciousness (5)	0.026
B4: Spatiotemporal grain that matters for consciousness is the one that maximizes phi (8)	0.026
B3: Functional (extrinsic) equivalence doesn't entail phenomenal (intrinsic) equivalence (8)	0.026
D7: Changes in the strength of connections within PSC should be associated with changes in experience, even if neural activity does not change (8)	0.026
C8: Only distinctions with c-e powers compatible with the c-e powers of the whole PSC contribute to C (8)	0.026
E10: There should be content-specific synchronization between high-level and low-level areas in the posterior cortex, regardless of the task (12; 13)	0.000
E2: Local strengthening or weakening of horizontal connections in topographic areas should lead to a local distortion of experienced visual space (10)	0.000
E8: Expert meditators experiencing 'pure presence' should exhibit decrease in gamma band, especially in posterior areas (11)	0.000
E6: Deactivating already inactive neurons of retinotopic cortex should remove part of spatial experience (10)	0.000
F1: Conscious perception overflows report (1; 14)	0.000
F3: Artificial systems, even if functionally equivalent to human brains, won't be conscious (no high PHI) (8)	0.000
E14: Loss and recovery of consciousness should be associated with the breakdown and recovery of the capacity for information integration. (5)	0.000

E16: Across various possible spatiotemporal grains to study the brain, the one constitutive of consciousness should be the one achieving maximal phi (3)	0.000
E13: When c-e powers of neuronal populations are incongruent with c-e powers of the PSC, corresponding stimulus should remain unconscious (9)	0.000
E11: Similarity metrics should be higher for content-specific activation than task-specific activation in some posterior areas (12)	0.000
E12: Similarity judgments between experiences should map onto similarity between CESs, as reconstructed from brain states (2)	0.000
F2: Because mental action is not essential to consciousness, in a MIB task actively looking for a stimulus should not facilitate conscious access (10)	0.000
E5: Artificial activation of neurons near the then terminus in a directed grid should result in perceiving a stimulus as having occurred earlier (15)	0.000
E4: Modulation of synaptic strength/excitability of neurons in directed grids should change properties of phenomenal flow regardless of activity levels (15)	0.000
E3: Duration of the extended present should be proportional to the number of neurons constituting a directed grid (15)	0.000
E1: All features of a consciously perceived stimulus should be maximally decodable from posterior areas, regardless of task (4; 12; 13)	0.000
E15: Reduction of transmission in callosal fibres should lead to a point at which one consciousness splits in two. (Split brain) (1)	0.000

Table S1.6. *IIT Harmonic Centrality.*

Claim	BC
C2: Type of experience (e.g., spatial vs temporal) should depend on the architecture of the substrate specifying the CES (1)	0.024
B2: ACTIVITY: changes in the state of the PSC units determine changes in CES (and therefore in experience) (8)	0.020
C3: Entire PSC (not only task-relevant activity) supports CES (= experience) (1)	0.020
C9: PSC needs to be both specialized and integrated to achieve high phi (5)	0.013
D2: Temporal structure of experience should map into features specified by neurons organized in directed grids (15)	0.012
D1: Posterior cortex, which has the right architecture for high integration and specialization, is the likely candidate for NCC (4)	0.010
D3: Spatial structure of experience should map into features specified by neurons organized in non-directed grid-like networks (6)	0.009
D5: Inactive neurons can contribute to consciousness (1)	0.008
C5: All units in the PSC must have the potential to be active (1)	0.007

D4: Horizontal connections within topographically organized visual areas would be needed to experience visual space (6; 7)	0.007
E7: Experience of 'pure presence' should exhibit largely inactive posterior cortex (11)	0.005
C7: Similarity between PHI structures should map into similarity between experiences (8)	0.004
B1: CONNECTIVITY: the connections between units of a PSC determine whether it can support high phi (8)	0.003
D6: Brain Complexity measures sensitive to specialization / integration should correlate with consciousness (5)	0.003
E9: Network that corresponds to content of C should be actively maintained for the duration of that content in C (13)	0.001
D7: Changes in the strength of connections within PSC should be associated with changes in experience, even if neural activity does not change (8)	0.001
C8: Only distinctions with c-e powers compatible with the c-e powers of the whole PSC contribute to C (8)	0.001
B4: Spatiotemporal grain that matters for consciousness is the one that maximizes phi (8)	0.001
B3: Functional (extrinsic) equivalence doesn't entail phenomenal (intrinsic) equivalence (8)	0.001
Core: Based on introspection, every experience is essentially intrinsic, specific, unitary, definite, and structured = Operationally, the PSC must be a maximum of irreducible, specific, intrinsic c-e power, supporting a CES (1; 8)	0.000
C1: Lattices of specialized + integrated units are well-suited to achieve high phi (7)	0.000
C6: Every distinction that is consciously experienced should be manifested in the CES for as long as it is experienced (8; 12; 13)	0.000
C4: C is extremely rich (i.e., highly structured) (14)	0.000
E10: There should be content-specific synchronization between high-level and low-level areas in the posterior cortex, regardless of the task (12; 13)	0.000
E2: Local strengthening or weakening of horizontal connections in topographic areas should lead to a local distortion of experienced visual space (10)	0.000
E8: Expert meditators experiencing 'pure presence' should exhibit decrease in gamma band, especially in posterior areas (11)	0.000
E6: Deactivating already inactive neurons of retinotopic cortex should remove part of spatial experience (10)	0.000
F1: Conscious perception overflows report (1; 14)	0.000
F3: Artificial systems, even if functionally equivalent to human brains, won't be conscious (no high PHI) (8)	0.000

E14: Loss and recovery of consciousness should be associated with the breakdown and recovery of the capacity for information integration. (5)	0.000
E16: Across various possible spatiotemporal grains to study the brain, the one constitutive of consciousness should be the one achieving maximal phi (3)	0.000
E13: When c-e powers of neuronal populations are incongruent with c-e powers of the PSC, corresponding stimulus should remain unconscious (9)	0.000
E11: Similarity metrics should be higher for content-specific activation than task-specific activation in some posterior areas (12)	0.000
E12: Similarity judgments between experiences should map onto similarity between CESs, as reconstructed from brain states (2)	0.000
F2: Because mental action is not essential to consciousness, in a MIB task actively looking for a stimulus should not facilitate conscious access (10)	0.000
E5: Artificial activation of neurons near the then terminus in a directed grid should result in perceiving a stimulus as having occurred earlier (15)	0.000
E4: Modulation of synaptic strength/excitability of neurons in directed grids should change properties of phenomenal flow regardless of activity levels (15)	0.000
E3: Duration of the extended present should be proportional to the number of neurons constituting a directed grid (15)	0.000
E1: All features of a consciously perceived stimulus should be maximally decodable from posterior areas, regardless of task (4; 12; 13)	0.000
E15: Reduction of transmission in callosal fibres should lead to a point at which one consciousness splits in two. (Split brain) (1)	0.000

Table S1.7. *IIT Betweenness Centrality.*

Claim	PR
E10: There should be content-specific synchronization between high-level and low-level areas in the posterior cortex, regardless of the task (12; 13)	0.049
E2: Local strengthening or weakening of horizontal connections in topographic areas should lead to a local distortion of experienced visual space (10)	0.045
E8: Expert meditators experiencing 'pure presence' should exhibit decrease in gamma band, especially in posterior areas (11)	0.038
E6: Deactivating already inactive neurons of retinotopic cortex should remove part of spatial experience (10)	0.038
F1: Conscious perception overflows report (1; 14)	0.037
D4: Horizontal connections within topographically organized visual areas would be needed to experience visual space (6; 7)	0.037
F3: Artificial systems, even if functionally equivalent to human brains, won't be conscious (no high PHI) (8)	0.034

E9: Network that corresponds to content of C should be actively maintained for the duration of that content in C (13)	0.031
E14: Loss and recovery of consciousness should be associated with the breakdown and recovery of the capacity for information integration. (5)	0.030
E16: Across various possible spatiotemporal grains to study the brain, the one constitutive of consciousness should be the one achieving maximal phi (3)	0.029
D5: Inactive neurons can contribute to consciousness (1)	0.029
E13: When c-e powers of neuronal populations are incongruent with c-e powers of the PSC, corresponding stimulus should remain unconscious (9)	0.029
E11: Similarity metrics should be higher for content-specific activation than task-specific activation in some posterior areas (12)	0.028
E7: Experience of 'pure presence' should exhibit largely inactive posterior cortex (11)	0.028
C2: Type of experience (e.g., spatial vs temporal) should depend on the architecture of the substrate specifying the CES (1)	0.027
D2: Temporal structure of experience should map into features specified by neurons organized in directed grids (15)	0.026
D3: Spatial structure of experience should map into features specified by neurons organized in non-directed grid-like networks (6)	0.026
D1: Posterior cortex, which has the right architecture for high integration and specialization, is the likely candidate for NCC (4)	0.026
E12: Similarity judgments between experiences should map onto similarity between CESs, as reconstructed from brain states (2)	0.023
F2: Because mental action is not essential to consciousness, in a MIB task actively looking for a stimulus should not facilitate conscious access (10)	0.023
E5: Artificial activation of neurons near the then terminus in a directed grid should result in perceiving a stimulus as having occurred earlier (15)	0.022
E4: Modulation of synaptic strength/excitability of neurons in directed grids should change properties of phenomenal flow regardless of activity levels (15)	0.022
E3: Duration of the extended present should be proportional to the number of neurons constituting a directed grid (15)	0.022
E1: All features of a consciously perceived stimulus should be maximally decodable from posterior areas, regardless of task (4; 12; 13)	0.020
C4: C is extremely rich (i.e., highly structured) (14)	0.020
C3: Entire PSC (not only task-relevant activity) supports CES (= experience) (1)	0.019
C9: PSC needs to be both specialized and integrated to achieve high phi (5)	0.019
C7: Similarity between PHI structures should map into similarity between experiences (8)	0.019

D6: Brain Complexity measures sensitive to specialization / integration should correlate with consciousness (5)	0.019
E15: Reduction of transmission in callosal fibres should lead to a point at which one consciousness splits in two. (Split brain) (1)	0.019
B2: ACTIVITY: changes in the state of the PSC units determine changes in CES (and therefore in experience) (8)	0.018
B1: CONNECTIVITY: the connections between units of a PSC determine whether it can support high phi (8)	0.018
B4: Spatiotemporal grain that matters for consciousness is the one that maximizes phi (8)	0.018
B3: Functional (extrinsic) equivalence doesn't entail phenomenal (intrinsic) equivalence (8)	0.018
D7: Changes in the strength of connections within PSC should be associated with changes in experience, even if neural activity does not change (8)	0.017
C1: Lattices of specialized + integrated units are well-suited to achieve high phi (7)	0.017
C5: All units in the PSC must have the potential to be active (1)	0.017
C8: Only distinctions with c-e powers compatible with the c-e powers of the whole PSC contribute to C (8)	0.017
C6: Every distinction that is consciously experienced should be manifested in the CES for as long as it is experienced (8; 12; 13)	0.017
Core: Based on introspection, every experience is essentially intrinsic, specific, unitary, definite, and structured = Operationally, the PSC must be a maximum of irreducible, specific, intrinsic c-e power, supporting a CES (1; 8)	0.015

Table S1.8. *IIT PageRank.*

PRM

Claim	LRC
Core: Given background conditions, a first-order state is conscious if it is indexed, via a pointer, by a higher-order state, as reliably reflecting the world as it is now (5; 8)	1.000
B2: C requires higher-order cognitive states (5)	0.556
B1: The function of the pointer (i.e., inner awareness) is an implicit metacognitive ability (1)	0.500
B4: HOS can incorrectly index the reliability of the FOS (5)	0.278
C6: Indexing reliability of FOS incorrectly leads to inflation (1)	0.222
C2: Inner awareness must discriminate between neural activity due to external stimulus and noise (8)	0.194

C1: In humans, PFC, held to be involved in metacognitive abilities, is required for higher-order states (1; 5)	0.167
D2: A change in HOS should lead to a change in C even if the FOS remains the same (5)	0.167
B5: FOS represent the world, HOS track the reliability of FOS (1)	0.139
E1: In humans, PFC is necessary for conscious perception (2; 5)	0.111
F1: TMS stimulation of PFC and lesions should impact subjective reports of C (1; 5)	0.083
D3: Imagery vividness is modulated by whether the HOS labels an internally generated FOS as more similar to noise or to an external stimulus (1)	0.083
C9: Function of C is formation of subjectively justified beliefs about the external world (i.e., monitoring reality) (1)	0.083
C8: FOS drive task performance (1; 5)	0.083
D10: Given the right paradigm, discrimination performance should be the same with and without conscious perception (1)	0.056
C3: HOS relies on averages to track internal noise (8)	0.056
C7: Metacognitive capacities to track task performance are reduced (3)	0.056
D7: Inflation should be modulated by attention: when attention is reduced, participants should be more likely to say something is there when it is not (1; 4)	0.056
B3: If you are not aware of being in a mental state, then there's nothing it is like for you to be in that mental state (1)	0.056
D8: Increased tendency to make false alarms in detection tasks for stimuli presented peripherally (3)	0.028
F2: Manipulating the PFC should induce confusion between perception and imagery (1)	0.028
E2: System tracking reliability of sensory signal might not track updated baseline neuronal excitability (8)	0.028
C4: Conscious states need to be accessed (no P-c without A-c) (4; 5)	0.028
C5: Subjects believe that C is richer than it actually is (4; 5)	0.028
D6: Inflation should be modulated by expectations: inflation is stronger when expectations are stronger (4)	0.028
D12: Inner awareness should correlate with implicit metacognitive abilities (1)	0.028
F8: Impairing the functioning of DLPFC (e.g., with TMS) should change subjective report about awareness without affecting task performance (1)	0.000
F3: Visual hallucinations can occur if functions of PFC are disrupted (e.g., Schizophrenia, dopaminergic drugs, direct cortical stimulation) (1)	0.000

F7: Decodability of inflated perception should be high in later stages of the visual processing hierarchy, not in primary visual cortex (4)	0.000
F5: Stimuli presented when neuronal excitability is high should lead to more false alarms and higher visibility/confidence ratings (8)	0.000
D4: Perception doesn't overflow reportability (4)	0.000
D11: Adaptation should affect reality monitoring system, leading to shifts in awareness without changes in discrimination performance (9)	0.000
F6: Top-down effects of expectations on inflation should be reflected by timecourses of decodability between frontoparietal and visual areas (4)	0.000
D1: Given background conditions, artificial and biological systems with FOS, higher-order mechanisms, and metacognitive abilities are conscious (6; 7)	0.000
D5: Participants should report rich phenomenology while missing details of the percept (3)	0.000
F4: Cross-decoding of neural populations responsible for perceptual and imagery vividness should be weak or absent (9)	0.000
D9: Phenomenological control should be able to alter reliability of FOS, but not their content (9)	0.000

Table S1.9. *PRM Local Reach Centrality. Claims in grey are the ones tested by the ETHOS adversarial collaboration. Abbreviations and references are reported in the main text.*

Claim	HC
Core: Given background conditions, a first-order state is conscious if it is indexed, via a pointer, by a higher-order state, as reliably reflecting the world as it is now (5; 8)	0.460
B2: C requires higher-order cognitive states (5)	0.288
B1: The function of the pointer (i.e., inner awareness) is an implicit metacognitive ability (1)	0.275
C6: Indexing reliability of FOS incorrectly leads to inflation (1)	0.162
B4: HOS can incorrectly index the reliability of the FOS (5)	0.146
C1: In humans, PFC, held to be involved in metacognitive abilities, is required for higher-order states (1; 5)	0.134
B5: FOS represent the world, HOS track the reliability of FOS (1)	0.125
C2: Inner awareness must discriminate between neural activity due to external stimulus and noise (8)	0.120
D2: A change in HOS should lead to a change in C even if the FOS remains the same (5)	0.111
D3: Imagery vividness is modulated by whether the HOS labels an internally generated FOS as more similar to noise or to an external stimulus (1)	0.083

E1: In humans, PFC is necessary for conscious perception (2; 5)	0.079
F1: TMS stimulation of PFC and lesions should impact subjective reports of C (1; 5)	0.069
C9: Function of C is formation of subjectively justified beliefs about the external world (i.e., monitoring reality) (1)	0.056
C8: FOS drive task performance (1; 5)	0.056
D10: Given the right paradigm, discrimination performance should be the same with and without conscious perception (1)	0.056
C3: HOS relies on averages to track internal noise (8)	0.042
C7: Metacognitive capacities to track task performance are reduced (3)	0.042
D7: Inflation should be modulated by attention: when attention is reduced, participants should be more likely to say something is there when it is not (1; 4)	0.042
B3: If you are not aware of being in a mental state, then there's nothing it is like for you to be in that mental state (1)	0.042
D8: Increased tendency to make false alarms in detection tasks for stimuli presented peripherally (3)	0.028
F2: Manipulating the PFC should induce confusion between perception and imagery (1)	0.028
E2: System tracking reliability of sensory signal might not track updated baseline neuronal excitability (8)	0.028
C4: Conscious states need to be accessed (no P-c without A-c) (4; 5)	0.028
C5: Subjects believe that C is richer than it actually is (4; 5)	0.028
D6: Inflation should be modulated by expectations: inflation is stronger when expectations are stronger (4)	0.028
D12: Inner awareness should correlate with implicit metacognitive abilities (1)	0.028
F8: Impairing the functioning of DLPFC (e.g., with TMS) should change subjective report about awareness without affecting task performance (1)	0.000
F3: Visual hallucinations can occur if functions of PFC are disrupted (e.g., Schizophrenia, dopaminergic drugs, direct cortical stimulation) (1)	0.000
F7: Decodability of inflated perception should be high in later stages of the visual processing hierarchy, not in primary visual cortex (4)	0.000
F5: Stimuli presented when neuronal excitability is high should lead to more false alarms and higher visibility/confidence ratings (8)	0.000
D4: Perception doesn't overflow reportability (4)	0.000
D11: Adaptation should affect reality monitoring system, leading to shifts in awareness without changes in discrimination performance (9)	0.000

F6: Top-down effects of expectations on inflation should be reflected by timecourses of decodability between frontoparietal and visual areas (4)	0.000
D1: Given background conditions, artificial and biological systems with FOS, higher-order mechanisms, and metacognitive abilities are conscious (6; 7)	0.000
D5: Participants should report rich phenomenology while missing details of the percept (3)	0.000
F4: Cross-decoding of neural populations responsible for perceptual and imagery vividness should be weak or absent (9)	0.000
D9: Phenomenological control should be able to alter reliability of FOS, but not their content (9)	0.000

Table S1.10. *PRM Harmonic Centrality.*

Claim	BC
B1: The function of the pointer (i.e., inner awareness) is an implicit metacognitive ability (1)	0.014
C6: Indexing reliability of FOS incorrectly leads to inflation (1)	0.012
C2: Inner awareness must discriminate between neural activity due to external stimulus and noise (8)	0.011
D3: Imagery vividness is modulated by whether the HOS labels an internally generated FOS as more similar to noise or to an external stimulus (1)	0.010
C1: In humans, PFC, held to be involved in metacognitive abilities, is required for higher-order states (1; 5)	0.009
B4: HOS can incorrectly index the reliability of the FOS (5)	0.007
C3: HOS relies on averages to track internal noise (8)	0.006
F1: TMS stimulation of PFC and lesions should impact subjective reports of C (1; 5)	0.006
D2: A change in HOS should lead to a change in C even if the FOS remains the same (5)	0.005
B2: C requires higher-order cognitive states (5)	0.005
D10: Given the right paradigm, discrimination performance should be the same with and without conscious perception (1)	0.005
C9: Function of C is formation of subjectively justified beliefs about the external world (i.e., monitoring reality) (1)	0.004
D8: Increased tendency to make false alarms in detection tasks for stimuli presented peripherally (3)	0.004
E2: System tracking reliability of sensory signal might not track updated baseline neuronal excitability (8)	0.004
B5: FOS represent the world, HOS track the reliability of FOS (1)	0.003
C7: Metacognitive capacities to track task performance are reduced (3)	0.002

D7: Inflation should be modulated by attention: when attention is reduced, participants should be more likely to say something is there when it is not (1; 4)	0.002
F2: Manipulating the PFC should induce confusion between perception and imagery (1)	0.002
C5: Subjects believe that C is richer than it actually is (4; 5)	0.002
B3: If you are not aware of being in a mental state, then there's nothing it is like for you to be in that mental state (1)	0.002
C4: Conscious states need to be accessed (no P-c without A-c) (4; 5)	0.002
D6: Inflation should be modulated by expectations: inflation is stronger when expectations are stronger (4)	0.002
D12: Inner awareness should correlate with implicit metacognitive abilities (1)	0.001
C8: FOS drive task performance (1; 5)	0.001
Core: Given background conditions, a first-order state is conscious if it is indexed, via a pointer, by a higher-order state, as reliably reflecting the world as it is now (5; 8)	0.000
E1: In humans, PFC is necessary for conscious perception (2; 5)	0.000
F8: Impairing the functioning of DLPFC (e.g., with TMS) should change subjective report about awareness without affecting task performance (1)	0.000
F3: Visual hallucinations can occur if functions of PFC are disrupted (e.g., Schizophrenia, dopaminergic drugs, direct cortical stimulation) (1)	0.000
F7: Decodability of inflated perception should be high in later stages of the visual processing hierarchy, not in primary visual cortex (4)	0.000
F5: Stimuli presented when neuronal excitability is high should lead to more false alarms and higher visibility/confidence ratings (8)	0.000
D4: Perception doesn't overflow reportability (4)	0.000
D11: Adaptation should affect reality monitoring system, leading to shifts in awareness without changes in discrimination performance (9)	0.000
F6: Top-down effects of expectations on inflation should be reflected by timecourses of decodability between frontoparietal and visual areas (4)	0.000
D1: Given background conditions, artificial and biological systems with FOS, higher-order mechanisms, and metacognitive abilities are conscious (6; 7)	0.000
D5: Participants should report rich phenomenology while missing details of the percept (3)	0.000
F4: Cross-decoding of neural populations responsible for perceptual and imagery vividness should be weak or absent (9)	0.000
D9: Phenomenological control should be able to alter reliability of FOS, but not their content (9)	0.000

Table S1.11. *PRM Betweenness Centrality.*

Claim	PR
F8: Impairing the functioning of DLPFC (e.g., with TMS) should change subjective report about awareness without affecting task performance (1)	0.063
F3: Visual hallucinations can occur if functions of PFC are disrupted (e.g., Schizophrenia, dopaminergic drugs, direct cortical stimulation) (1)	0.052
F7: Decodability of inflated perception should be high in later stages of the visual processing hierarchy, not in primary visual cortex (4)	0.051
D10: Given the right paradigm, discrimination performance should be the same with and without conscious perception (1)	0.046
D8: Increased tendency to make false alarms in detection tasks for stimuli presented peripherally (3)	0.044
F5: Stimuli presented when neuronal excitability is high should lead to more false alarms and higher visibility/confidence ratings (8)	0.040
D4: Perception doesn't overflow reportability (4)	0.037
D11: Adaptation should affect reality monitoring system, leading to shifts in awareness without changes in discrimination performance (9)	0.036
F2: Manipulating the PFC should induce confusion between perception and imagery (1)	0.036
F6: Top-down effects of expectations on inflation should be reflected by timecourses of decodability between frontoparietal and visual areas (4)	0.033
F1: TMS stimulation of PFC and lesions should impact subjective reports of C (1; 5)	0.033
D1: Given background conditions, artificial and biological systems with FOS, higher-order mechanisms, and metacognitive abilities are conscious (6; 7)	0.032
E2: System tracking reliability of sensory signal might not track updated baseline neuronal excitability (8)	0.032
D5: Participants should report rich phenomenology while missing details of the percept (3)	0.029
D3: Imagery vividness is modulated by whether the HOS labels an internally generated FOS as more similar to noise or to an external stimulus (1)	0.028
C4: Conscious states need to be accessed (no P-c without A-c) (4; 5)	0.027
F4: Cross-decoding of neural populations responsible for perceptual and imagery vividness should be weak or absent (9)	0.022
C3: HOS relies on averages to track internal noise (8)	0.021
C1: In humans, PFC, held to be involved in metacognitive abilities, is required for higher-order states (1; 5)	0.021

C9: Function of C is formation of subjectively justified beliefs about the external world (i.e., monitoring reality) (1)	0.021
C6: Indexing reliability of FOS incorrectly leads to inflation (1)	0.020
D9: Phenomenological control should be able to alter reliability of FOS, but not their content (9)	0.020
B1: The function of the pointer (i.e., inner awareness) is an implicit metacognitive ability (1)	0.019
E1: In humans, PFC is necessary for conscious perception (2; 5)	0.018
C7: Metacognitive capacities to track task performance are reduced (3)	0.018
D7: Inflation should be modulated by attention: when attention is reduced, participants should be more likely to say something is there when it is not (1; 4)	0.018
C5: Subjects believe that C is richer than it actually is (4; 5)	0.018
D6: Inflation should be modulated by expectations: inflation is stronger when expectations are stronger (4)	0.018
C2: Inner awareness must discriminate between neural activity due to external stimulus and noise (8)	0.018
D12: Inner awareness should correlate with implicit metacognitive abilities (1)	0.018
D2: A change in HOS should lead to a change in C even if the FOS remains the same (5)	0.017
C8: FOS drive task performance (1; 5)	0.017
B4: HOS can incorrectly index the reliability of the FOS (5)	0.016
B2: C requires higher-order cognitive states (5)	0.016
B5: FOS represent the world, HOS track the reliability of FOS (1)	0.016
B3: If you are not aware of being in a mental state, then there's nothing it is like for you to be in that mental state (1)	0.016
Core: Given background conditions, a first-order state is conscious if it is indexed, via a pointer, by a higher-order state, as reliably reflecting the world as it is now (5; 8)	0.014

Table S1.12. *PRM PageRank.*

RPT

Claim	LRC
Core: Recurrent processing (RP) that changes network activity in sensory areas and enables perceptual organization is both necessary and sufficient for consciousness (2)	1.000
B5: Every function that requires RP that changes network activity will only be possible during conscious perception (3)	0.483

B1: Plasticity (potential for neuronal activity to change connectivity) is needed for consciousness (2)	0.310
B4: Information in visual short-term memory, which requires RP, is conscious (3)	0.241
E1: NMDA receptors, which enable cortico-cortical feedback and synaptic plasticity, are needed for consciousness (6)	0.241
B2: System with purely FF architecture cannot be conscious (3)	0.241
C5: Attention and availability for report are not necessary for C (4)	0.207
C1: Recurrency creates conditions for Hebbian learning (2)	0.138
B3: Functions that are based on FF processing can be performed without consciousness (7)	0.138
C6: Perceptual organization requires RP that changes network activity: it's possible only during conscious perception (7)	0.138
C2: Only consciously perceived stimuli evoke RP and have lasting impact on the brain (2)	0.069
F1: NMDA receptors activation should be precondition for large-scale RP requiring consciousness (6)	0.034
E2: Involvement of frontal areas is not necessary for C (3)	0.034
D11: Figure-ground segregation should be possible only during conscious perception (7)	0.034
D12: Perceptual grouping (integration) should be possible only during conscious perception (7)	0.034
F4: Inhibiting RP in sensory areas should impair perceptual organization (7)	0.000
F2: Inhibiting large-scale RP by targeting NMDA receptors should result in loss of (state) consciousness (6)	0.000
D1: Given a discrepancy between conscious perception and physical features of the stimulus, learning should occur for the former (2)	0.000
F3: NCC should not involve the PFC (3)	0.000
D2: Artificial and biological systems with recurrent architecture that enables learning and perceptual organization are conscious (6; 8)	0.000
D10: Learning can take place for conscious but unattended events/stimuli (2)	0.000
D3: Unconscious learning should be impossible (2)	0.000
C4: Stimuli in change blindness and inattentional blindness paradigms are consciously perceived (3)	0.000
C3: Neural markers (RP) can serve as sufficient evidence for C, irrespective of report (5)	0.000
D8: Conscious perception overflows report, hence, perception is rich (5)	0.000
D5: Stimulus localization is possible even if the stimulus is unconsciously processed (7)	0.000

D6: Base groupings/feature conjunction (e.g. combination of colour and orientation) are possible if the stimulus is unconsciously processed (1)	0.000
D7: Some interference (e.g., grey dots of different luminosity appearing on bright/dark background) could be possible even if stimulus is unconscious (7)	0.000
D4: Feature extraction and higher level categorizations (shape, object category) are possible even if the stimulus is unconsciously processed (7)	0.000
D9: Some inference (like Kanizsa illusion) is possible only during conscious perception (7)	0.000

Table S1.13. *RPT Local Reach Centrality. Abbreviations and references are reported in the main text*

Claim	HC
Core: Recurrent processing (RP) that changes network activity in sensory areas and enables perceptual organization is both necessary and sufficient for consciousness (2)	0.517
B5: Every function that requires RP that changes network activity will only be possible during conscious perception (3)	0.273
B1: Plasticity (potential for neuronal activity to change connectivity) is needed for consciousness (2)	0.218
C5: Attention and availability for report are not necessary for C (4)	0.190
B2: System with purely FF architecture cannot be conscious (3)	0.172
E1: NMDA receptors, which enable cortico-cortical feedback and synaptic plasticity, are needed for consciousness (6)	0.149
B3: Functions that are based on FF processing can be performed without consciousness (7)	0.138
B4: Information in visual short-term memory, which requires RP, is conscious (3)	0.132
C1: Recurrency creates conditions for Hebbian learning (2)	0.121
C6: Perceptual organization requires RP that changes network activity: it's possible only during conscious perception (7)	0.121
C2: Only consciously perceived stimuli evoke RP and have lasting impact on the brain (2)	0.069
F1: NMDA receptors activation should be precondition for large-scale RP requiring consciousness (6)	0.034
E2: Involvement of frontal areas is not necessary for C (3)	0.034
D11: Figure-ground segregation should be possible only during conscious perception (7)	0.034
D12: Perceptual grouping (integration) should be possible only during conscious perception (7)	0.034
F4: Inhibiting RP in sensory areas should impair perceptual organization (7)	0.000

F2: Inhibiting large-scale RP by targeting NMDA receptors should result in loss of (state) consciousness (6)	0.000
D1: Given a discrepancy between conscious perception and physical features of the stimulus, learning should occur for the former (2)	0.000
F3: NCC should not involve the PFC (3)	0.000
D2: Artificial and biological systems with recurrent architecture that enables learning and perceptual organization are conscious (6; 8)	0.000
D10: Learning can take place for conscious but unattended events/stimuli (2)	0.000
D3: Unconscious learning should be impossible (2)	0.000
C4: Stimuli in change blindness and inattentional blindness paradigms are consciously perceived (3)	0.000
C3: Neural markers (RP) can serve as sufficient evidence for C, irrespective of report (5)	0.000
D8: Conscious perception overflows report, hence, perception is rich (5)	0.000
D5: Stimulus localization is possible even if the stimulus is unconsciously processed (7)	0.000
D6: Base groupings/feature conjunction (e.g. combination of colour and orientation) are possible if the stimulus is unconsciously processed (1)	0.000
D7: Some interference (e.g., grey dots of different luminosity appearing on bright/dark background) could be possible even if stimulus is unconscious (7)	0.000
D4: Feature extraction and higher level categorizations (shape, object category) are possible even if the stimulus is unconsciously processed (7)	0.000
D9: Some inference (like Kanizsa illusion) is possible only during conscious perception (7)	0.000

Table S1.14. *RPT Harmonic Centrality.*

Claim	BC
C5: Attention and availability for report are not necessary for C (4)	0.019
B4: Information in visual short-term memory, which requires RP, is conscious (3)	0.014
B3: Functions that are based on FF processing can be performed without consciousness (7)	0.010
B1: Plasticity (potential for neuronal activity to change connectivity) is needed for consciousness (2)	0.009
C6: Perceptual organization requires RP that changes network activity: it's possible only during conscious perception (7)	0.009
C1: Recurrency creates conditions for Hebbian learning (2)	0.008
B5: Every function that requires RP that changes network activity will only be possible during conscious perception (3)	0.007

C2: Only consciously perceived stimuli evoke RP and have lasting impact on the brain (2)	0.006
E2: Involvement of frontal areas is not necessary for C (3)	0.005
E1: NMDA receptors, which enable cortico-cortical feedback and synaptic plasticity, are needed for consciousness (6)	0.004
F1: NMDA receptors activation should be precondition for large-scale RP requiring consciousness (6)	0.002
B2: System with purely FF architecture cannot be conscious (3)	0.002
D11: Figure-ground segregation should be possible only during conscious perception (7)	0.002
D12: Perceptual grouping (integration) should be possible only during conscious perception (7)	0.002
Core: Recurrent processing (RP) that changes network activity in sensory areas and enables perceptual organization is both necessary and sufficient for consciousness (2)	0.000
F4: Inhibiting RP in sensory areas should impair perceptual organization (7)	0.000
F2: Inhibiting large-scale RP by targeting NMDA receptors should result in loss of (state) consciousness (6)	0.000
D1: Given a discrepancy between conscious perception and physical features of the stimulus, learning should occur for the former (2)	0.000
F3: NCC should not involve the PFC (3)	0.000
D2: Artificial and biological systems with recurrent architecture that enables learning and perceptual organization are conscious (6; 8)	0.000
D10: Learning can take place for conscious but unattended events/stimuli (2)	0.000
D3: Unconscious learning should be impossible (2)	0.000
C4: Stimuli in change blindness and inattentional blindness paradigms are consciously perceived (3)	0.000
C3: Neural markers (RP) can serve as sufficient evidence for C, irrespective of report (5)	0.000
D8: Conscious perception overflows report, hence, perception is rich (5)	0.000
D5: Stimulus localization is possible even if the stimulus is unconsciously processed (7)	0.000
D6: Base groupings/feature conjunction (e.g. combination of colour and orientation) are possible if the stimulus is unconsciously processed (1)	0.000
D7: Some interference (e.g., grey dots of different luminosity appearing on bright/dark background) could be possible even if stimulus is unconscious (7)	0.000
D4: Feature extraction and higher level categorizations (shape, object category) are possible even if the stimulus is unconsciously processed (7)	0.000
D9: Some inference (like Kanizsa illusion) is possible only during conscious perception (7)	0.000

Table S1.15. *RPT Betweenness Centrality.*

Claim	PR
F4: Inhibiting RP in sensory areas should impair perceptual organization (7)	0.069
F2: Inhibiting large-scale RP by targeting NMDA receptors should result in loss of (state) consciousness (6)	0.055
D1: Given a discrepancy between conscious perception and physical features of the stimulus, learning should occur for the former (2)	0.047
C5: Attention and availability for report are not necessary for C (4)	0.046
F3: NCC should not involve the PFC (3)	0.045
D2: Artificial and biological systems with recurrent architecture that enables learning and perceptual organization are conscious (6; 8)	0.040
D10: Learning can take place for conscious but unattended events/stimuli (2)	0.039
C1: Recurrency creates conditions for Hebbian learning (2)	0.037
C2: Only consciously perceived stimuli evoke RP and have lasting impact on the brain (2)	0.036
D3: Unconscious learning should be impossible (2)	0.036
C4: Stimuli in change blindness and inattentional blindness paradigms are consciously perceived (3)	0.034
F1: NMDA receptors activation should be precondition for large-scale RP requiring consciousness (6)	0.032
B3: Functions that are based on FF processing can be performed without consciousness (7)	0.031
B4: Information in visual short-term memory, which requires RP, is conscious (3)	0.030
E2: Involvement of frontal areas is not necessary for C (3)	0.029
C3: Neural markers (RP) can serve as sufficient evidence for C, irrespective of report (5)	0.029
D8: Conscious perception overflows report, hence, perception is rich (5)	0.029
D11: Figure-ground segregation should be possible only during conscious perception (7)	0.028
D12: Perceptual grouping (integration) should be possible only during conscious perception (7)	0.028
D5: Stimulus localization is possible even if the stimulus is unconsciously processed (7)	0.027
D6: Base groupings/feature conjunction (e.g. combination of colour and orientation) are possible if the stimulus is unconsciously processed (1)	0.027
D7: Some interference (e.g., grey dots of different luminosity appearing on bright/dark background) could be possible even if stimulus is unconscious (7)	0.027

D4: Feature extraction and higher level categorizations (shape, object category) are possible even if the stimulus is unconsciously processed (7)	0.027
C6: Perceptual organization requires RP that changes network activity: it's possible only during conscious perception (7)	0.026
E1: NMDA receptors, which enable cortico-cortical feedback and synaptic plasticity, are needed for consciousness (6)	0.026
D9: Some inference (like Kanizsa illusion) is possible only during conscious perception (7)	0.026
B1: Plasticity (potential for neuronal activity to change connectivity) is needed for consciousness (2)	0.024
B5: Every function that requires RP that changes network activity will only be possible during conscious perception (3)	0.024
B2: System with purely FF architecture cannot be conscious (3)	0.024
Core: Recurrent processing (RP) that changes network activity in sensory areas and enables perceptual organization is both necessary and sufficient for consciousness (2)	0.021

Table S1.16. *RPT PageRank.*

		LRC	**HC**	**BC**	**PR**
GNWT	90th percentile	0.179	0.126	0.006	0.033
	median	0.023	0.023	0.001	0.018
	10th percentile	0.000	0.000	0.000	0.015
IIT	90th percentile	0.292	0.176	0.012	0.037
	median	0.026	0.026	0.000	0.022
	10th percentile	0.000	0.000	0.000	0.017
PRM	90th percentile	0.244	0.152	0.009	0.045
	median	0.056	0.042	0.002	0.021
	10th percentile	0.000	0.000	0.000	0.016

RPT	90th percentile	0.248	0.193	0.009	0.046
	median	0.017	0.017	0.000	0.029
	10th percentile	0.000	0.000	0.000	0.024

Table S1.17. *Calculation of 90th percentile, median, and 10th percentile of each centrality measure.*

Supplementary S2: Centrality analysis for predictions tested by INTREPID and ETHOS

INTREPID and ETHOS are two consortia testing theories of consciousness via adversarial collaborations. INTREPID tests predictions of IIT and various theories inspired by predictive processing accounts (Olcese et al., 2024; more information in supplementary S3), while ETHOS tests adversarial hypotheses of different higher-order theories of consciousness (Fleming et al., 2025), including PRM. Here, we analyze the centrality of IIT's predictions tested by INTREPID and PRM's predictions tested by ETHOS.

IIT's tested predictions are **E2** ('Local strengthening or weakening of horizontal connections in topographic areas should lead to a local distortion of experienced visual space), **E6** ('Deactivating already inactive neurons of retinotopic cortex should remove part of spatial experience'), and **F2** ('Because mental action is not essential to consciousness, in a motion induced blindness task actively looking for a stimulus should not facilitate conscious access'). **E2** and **E6** reach a high PR value ($E2_{PR}$=0.045; $E6_{PR}$= 0.038), while **F2** has a medium value (0.023). This indicates that **E2** and **E6** are meaningful tests for IIT, whereas **F2** is much less informative for the theory.

Instead, the ETHOS experiments seem less significant for PRM: here, the relevant predictions are **D11** ('Adaptation should affect reality monitoring system, leading to shifts in awareness without changes in discrimination performance'), **F4** ('Cross-decoding of neural populations responsible for perceptual and imagery vividness should be weak or absent'), and **D9** ('Phenomenological control should be able to alter reliability of FOS, but not their content'). **D11** reaches a medium-high PR value (0.036), while **F4** and **D9** achieve only medium PR scores ($F4_{PR}$=0.022; $D9_{PR}$=0.020).

Supplementary S3: Additional theories

To better evaluate current attempts to adversarially test theories of consciousness, we created additional maps, in collaboration with theory proponents. Table S3.1 details the papers from which claims were extracted for each theory, and below we provide all resulting maps.

Theory	***Bibliography***
Active Inference (AI)	(1. Allen et al., 2016; 2. Friston et al., 2017; 3. Parr & Friston, 2018; 4. Parr, Corcoran, et al., 2019; 5. Smith et al., 2019; 6. Parr, Mirza, et al., 2019; 7. Smith et al., 2020; 8. Schlossmacher et al., 2020; 9. Whyte & Smith, 2021; 10. Smith et al., 2021; 11. Hesp et al., 2021; 12. Sandved-Smith et al., 2021; 13. Parr et al., 2022; 14. Whyte et al., 2022; 15. Olcese et al., 2024; 16. Rowe et al., 2024; 17. Hohwy, 2025; 18. Sandved-Smith et al., 2025; 19. Whyte et al., 2026; 20. Robinson et al., 2025)
Higher-order representation of a representation (HOROR)	(1. Brown, 2015, 2. 2025; 3. Fleming et al., 2025)
Higher-order state space (HOSS)	(1. Lau & Rosenthal, 2011; 2. Odegaard et al., 2017; 3. Brown et al., 2019; 4. Fleming, 2020; 5. Lau, 2022; 6. Butlin et al., 2023; 7. Michel, 2023; 8. Fleming & Michel, 2025; 9. Fleming et al., 2025)
Neurorepresentationalism (NREP)	(1. Pennartz, 2009, 2. 2015; 3. Olcese et al., 2018; 4. Pennartz, Dora, et al., 2019; 5. Pennartz, Farisco, et al., 2019; 6. Pennartz, 2022, 7. 2024; 8. Lee & Pennartz, 2026; 9. Corcoran et al., 2025)
Self-organizing metarepresentational account (SOMA)	(1. Strack et al., 1988; 2. Rosenthal, 1997; 3. Cleeremans et al., 2007; 4. Dienes & Perner, 2007; 5. Cleeremans, 2007; 6. Sirotin & Das, 2009; 7. Pasquali et al., 2010; 8. Cleeremans, 2011; 9. Lau & Rosenthal, 2011; 10. Timmermans et al., 2012; 11. Cleeremans, 2014; 12. Brown, 2015; 13. Cleermans, 2017; 14. Bulnes et al., 2019; 15. Cleeremans et al., 2020; 16. Peebles, 2022; 17. Cleeremans &

	Tallon-Baudry, 2022; 18. Lau, 2022; 19. Veit, 2023; 20. Shenhav, 2024; 21. Moncoucy et al., 2025; Fleming et al., 2025)

Table S3.1. *References for additional theories*

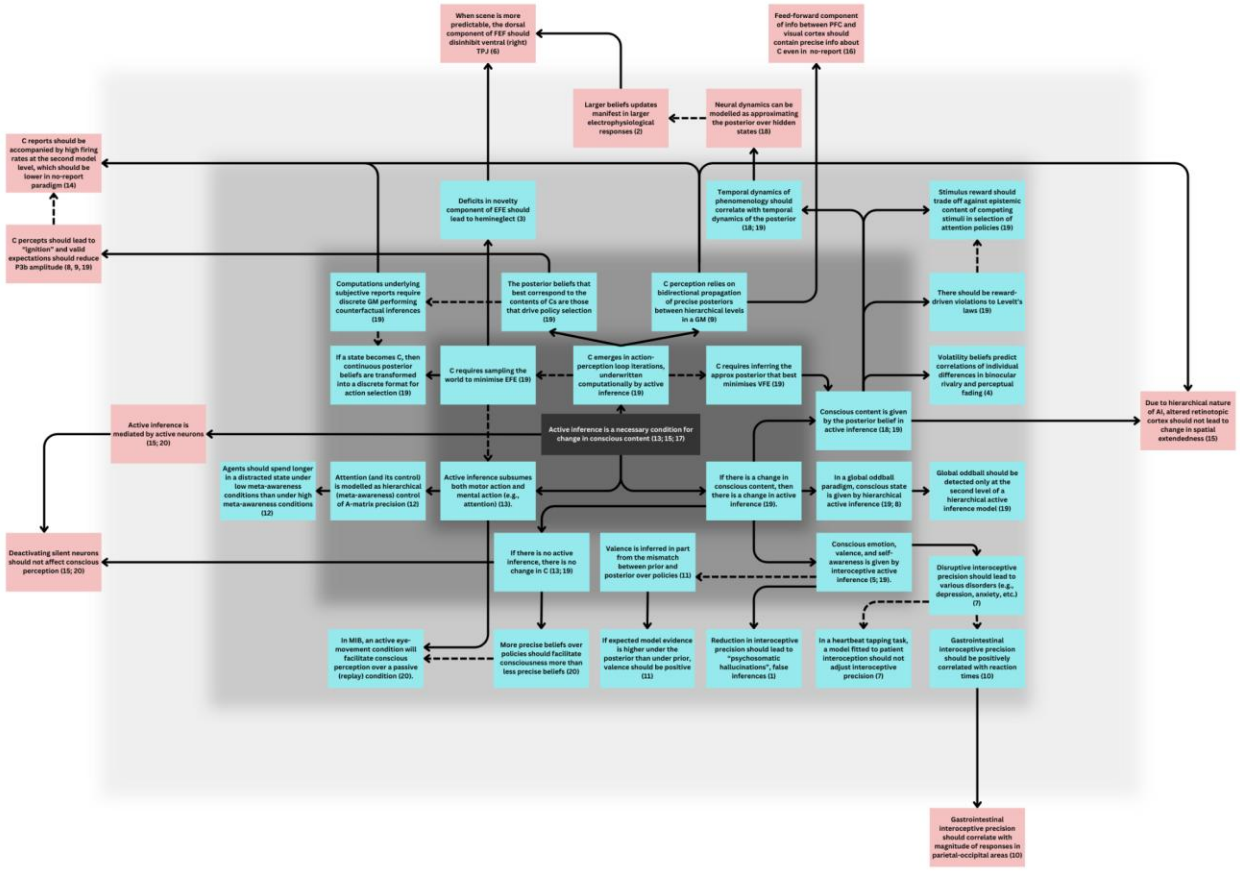


Figure S3.1. *Active Inference prediction map. Abbreviations: C (consciousness/conscious); EFE (expected free energy); FEF (frontal eye field); GM (generative model); MIB (motion-induced blindness); PFC (prefrontal cortex); TPJ (temporoparietal junction); VFE (variational free energy). See Table S3.1 for references.*

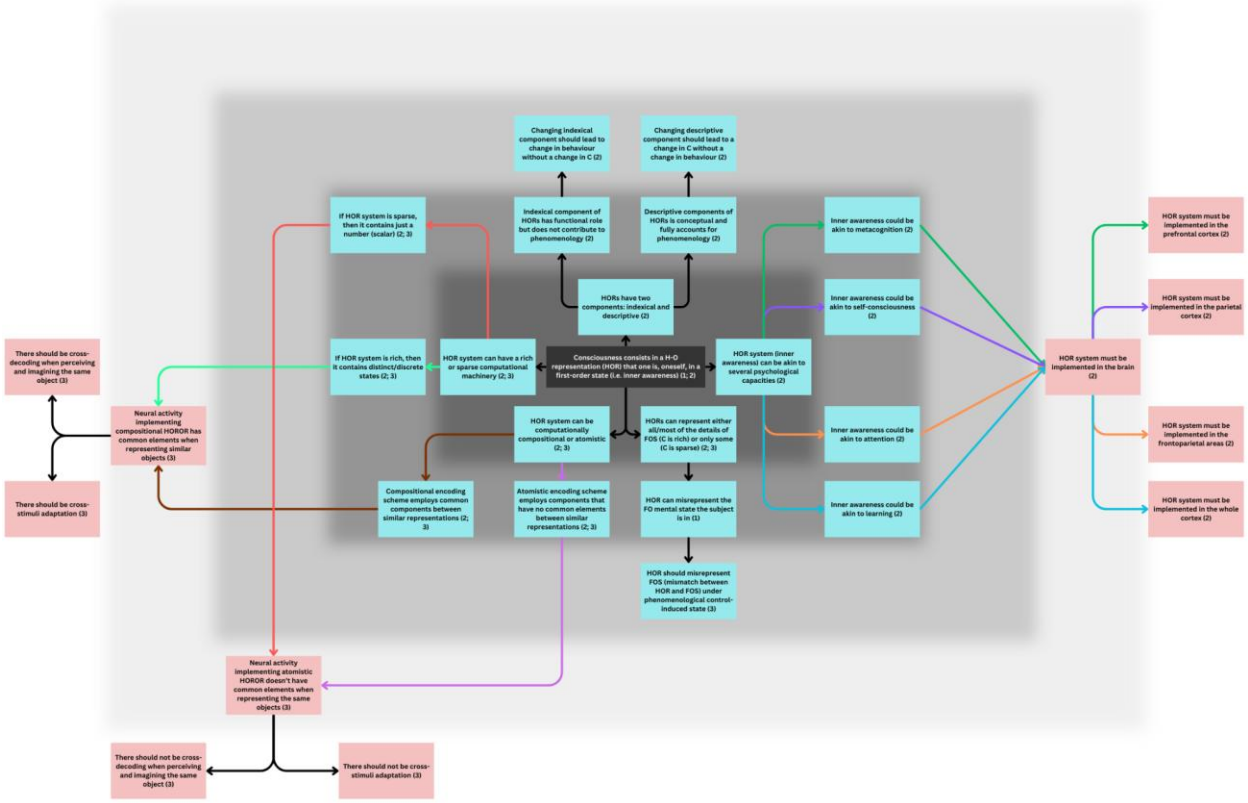


Figure S3.2. *HOROR prediction map. Arrows of different colors stemming from the same box represent mutually exclusive paths. Abbreviations: C (consciousness); FOS (first-order state); HOR (higher-order representation); HOROR (higher-order representation of a representation). See Table S3.1 for references.*

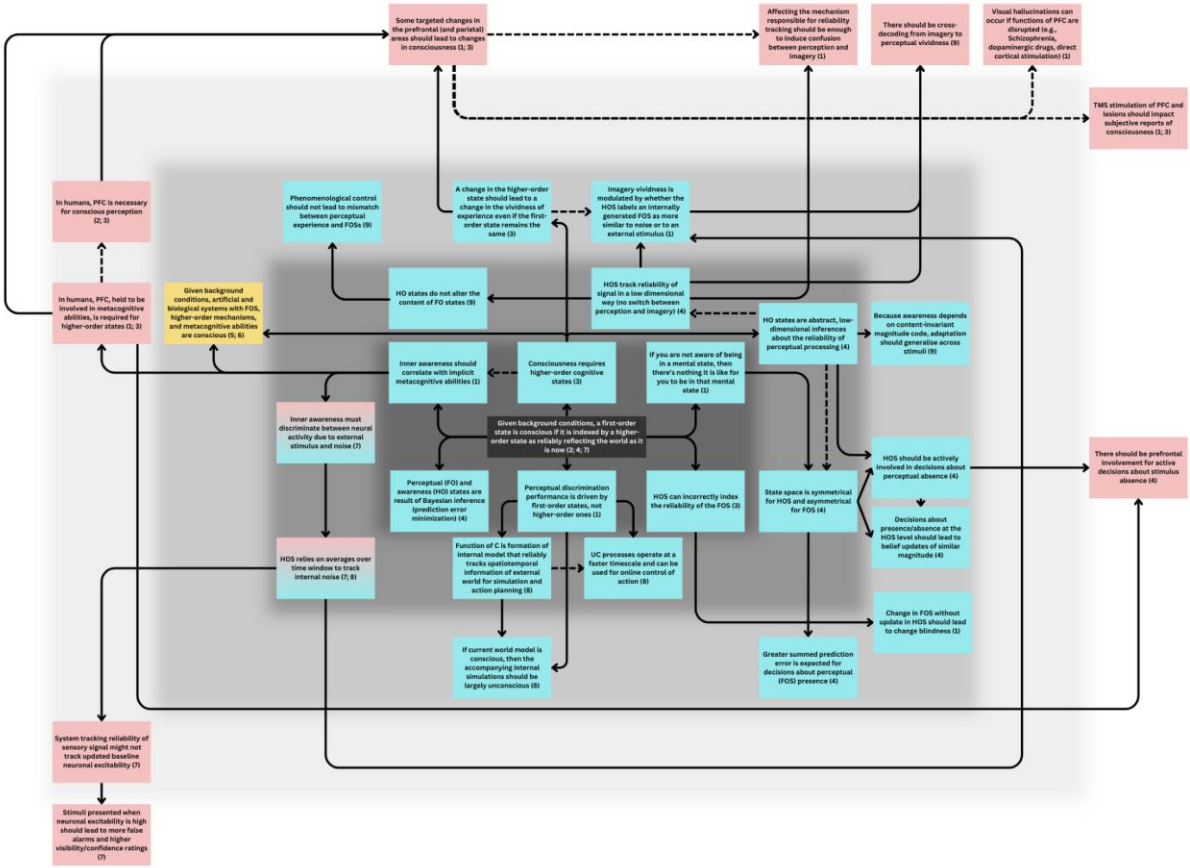


Figure S3.3. *HOSS prediction map. Abbreviations: C (consciousness); FOS (first-order state); HOS (higher-order state); PFC (prefrontal cortex); TMS (transcranial magnetic stimulation); UC (unconscious). See Table S3.1 for references.*

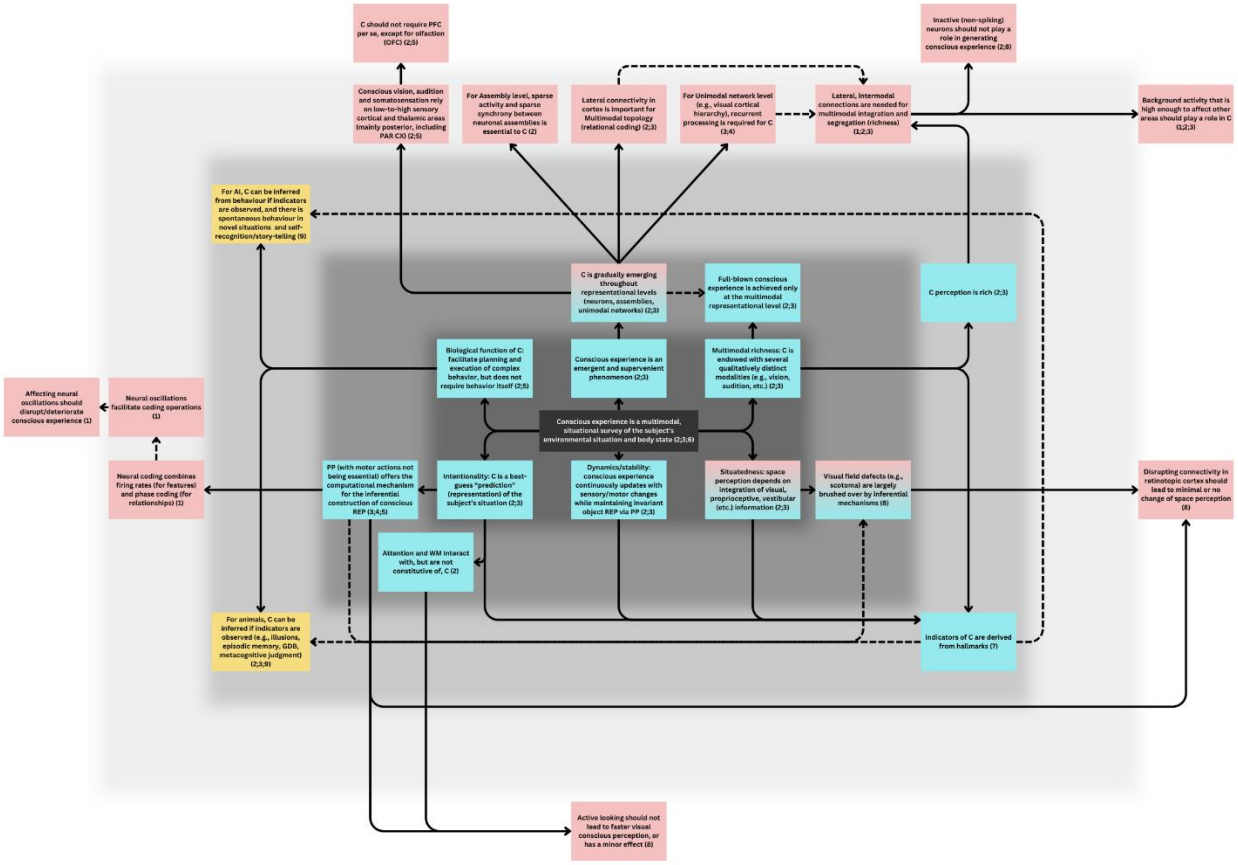


Figure S3.4. *NREP prediction map. Abbreviations: AI (artificial intelligence); C (consciousness); GDB (goal directed behavior); OFC (orbitofrontal cortex); PAR CX (parietal cortex); PFC (prefrontal cortex); PP (predictive processing); REP (representation); WM (working memory). See Table S3.1 for references.*

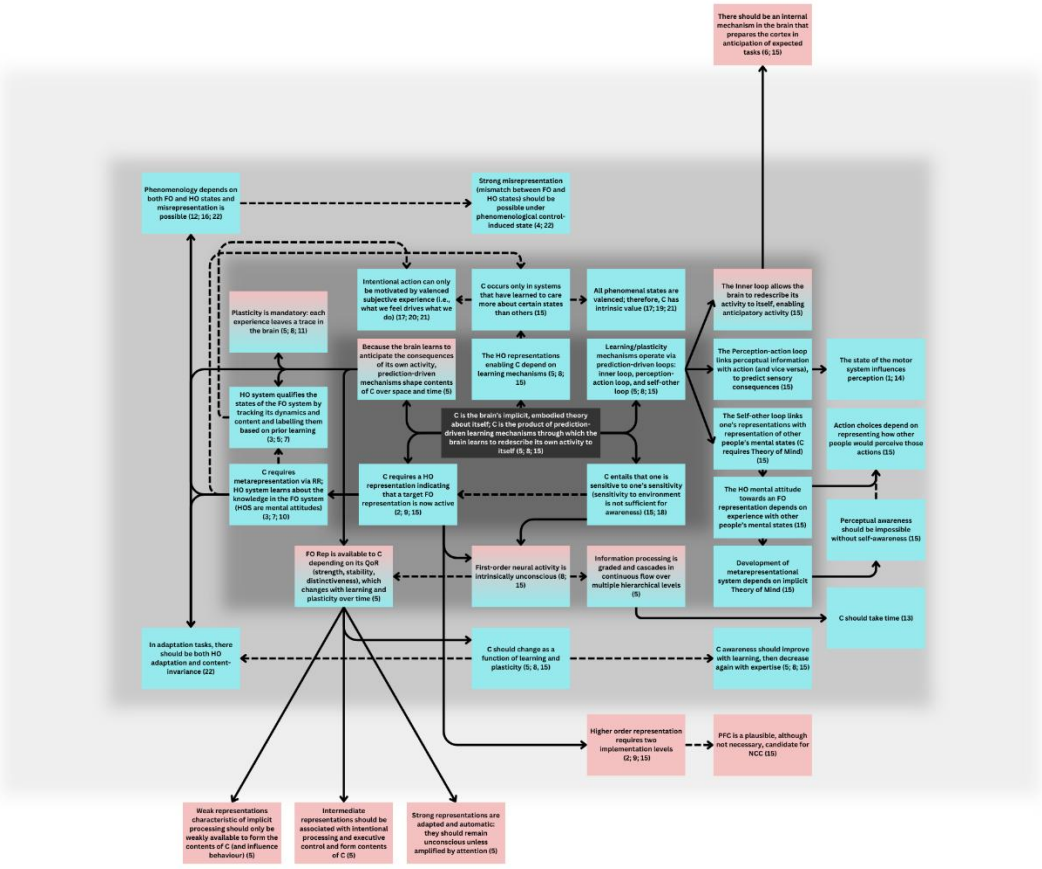


Figure S3.5. *SOMA prediction map. Abbreviations: C (consciousness), FO (first-order), HO (higher-order), HOS (higher-order state), NCC (neural correlates of consciousness), PFC (prefrontal cortex), QoR (quality of a representation), RR (representational redescription). See Table S3.1 for references.*

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